

The Information Costs of Climate Misinformation: Evidence from Global Financial Data

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Abstract

Climate misinformation can alter the information environment in which investors and liquidity providers trade. This paper examines that relationship in a firm-day panel of 204 listed climate-transition firms from 2012 to 2025. The primary outcome is the percent quoted bid–ask spread, a measure of market liquidity and an observable proxy for information asymmetry. Sector-specific exposure is constructed by classifying English-language Facebook climate posts against nine energy sub-industries. The estimates compare within-firm changes, with firm and year-month fixed effects, controls for trading and macroeconomic conditions, and standard errors clustered by firm and calendar date. Greater exposure is associated with significantly wider quoted spreads, and the association strengthens at a one-day lag. Price impact does not move in the opposite direction, while rolling CAPM beta and Fama–French excess returns remain weak under two-way clustering. This pattern is inconsistent with liquidity fragility and more consistent with adverse-selection pressure.

Keywords: Climate Misinformation, Information Asymmetry, Market Microstructure, Environmental Communication, Politics and Governance, Corporate Responsibility.

1 Introduction

The information landscape surrounding climate change has been systematically distorted by misinformation campaigns (Cobb, 2025). Markets aggregate information, but they do so at a price. Climate misinformation is produced for political ends—funded by fossil-fuel interests, amplified by ideologically motivated actors, and timed to political events (Farrell, 2016; Oreskes and Conway, 2010; Cinelli, De Francisci Morales, Galeazzi, Quattrociocchi, and Starnini, 2021)—yet market-microstructure theory gives two opposing predictions for its effect on trading costs. If misinformation increases the advantage of sophisticated traders who can distinguish credible signals from false ones, liquidity providers should widen bid–ask spreads to protect themselves. If misinformation instead attracts uninformed retail trading, the perceived share of informed order flow may fall and spreads may narrow even as the market’s capacity to absorb informed trading deteriorates. Which pattern prevails is the central question of this paper.

This question connects two related literatures. Political-behavior research shows that misinformation changes beliefs, trust, and collective action (Flynn, Nyhan, and Reifler, 2017; Bursztyrn, Rao, Roth, and Yanagizawa-Drott, 2020), while climate-finance research shows that climate news, disclosure, and policy uncertainty are priced in financial markets (Engle, Giglio, Kelly, Lee, and Stroebe, 2020; Bolton and Kacperczyk, 2021; Pankratz, Bauer, and Derwall, 2023). Neither literature traces the daily flow of politically produced climate misinformation into the mechanism that sets trading costs. Market microstructure provides that link because the quoted bid–ask spread is an observable price of information asymmetry and a direct measure of market liquidity (Glosten and Milgrom, 1985; Easley and O’Hara, 1992).

The theory distinguishes two causal channels. Under adverse selection, false claims about climate science, carbon policy, and the viability of low-carbon technologies make the implications for fossil-fuel and renewable firms harder to assess. Traders able to verify those claims gain an informational advantage over those who cannot. Liquidity providers then face greater expected losses when trading climate-transition equities with better-informed counterparties and widen spreads, while price impact rises. Under liquidity fragility, viral climate-policy and energy narratives instead draw noise traders into the affected sub-industries. A larger uninformed share lowers the perceived probability that the next order is informed, narrowing the spread, but shallow order flow allows each dollar traded to move prices

more, raising price impact. H1 tests the latter account: higher firm-day exposure to sector-specific climate misinformation should be associated with narrower percent quoted spreads and higher per-dollar price impact. Section 3 develops the mechanisms and their observable implications.

The empirical analysis uses 726,890 firm-day observations on 204 listed climate-transition firms from January 2012 to December 2025. The primary outcome is the percent quoted bid-ask spread; price impact supplies the second liquidity measure required for the joint test. Rolling CAPM beta and Fama-French five-factor excess returns serve as benchmark outcomes for systematic risk and return-pricing channels. The exposure measure is constructed by classifying 217,954 English-language Facebook climate posts from the Meta Content Library against nine TRBC energy sub-industries. Each firm inherits the exposure assigned to its sub-industry on each date, producing within-day cross-sectional variation while preserving the political and economic content of the posts.

Greater sector-specific misinformation exposure is associated with significantly wider quoted spreads after firm and year-month fixed effects, and the relationship strengthens at a one-day lag. Price impact does not display the divergence required by H1. The pattern is therefore inconsistent with liquidity fragility and more consistent with an adverse-selection response.

The paper makes three contributions. First, it extends market-microstructure theory to politically produced climate misinformation and distinguishes adverse selection from liquidity fragility through the joint behavior of spreads and price impact. Second, it provides a high-frequency firm-day exposure measure that maps political misinformation to the sub-industries whose values depend on the contested information. Third, it shows why inference requires within-day cross-sectional variation rather than a market-wide daily misinformation series. Together, these contributions connect the political production of misinformation to the liquidity of climate-transition equities without treating changes in risk or returns as the primary outcome.

The rest of the paper proceeds as follows. Section 2 reviews research on information and market liquidity and connects it to climate misinformation in financial markets. Section 3 develops the competing adverse-selection and liquidity-fragility mechanisms and states the hypothesis formally. Section 4 introduces the firm-day financial panel and misinformation corpus. Section 5 separates the market-liquidity design from the benchmark risk-and-return designs and sets out the identification and inference strategy. Section 6 reports the main estimates, placebo, and robustness checks. Section 7 interprets the findings, and Section 8 concludes.

2 Literature Review

2.1 Information and Market Liquidity

Market liquidity depends on the distribution and credibility of information. When some traders possess information that others lack, liquidity providers face the risk of trading at stale prices with better-informed counterparties (Akerlof, 1970; Glosten and Milgrom, 1985; Easley and O'Hara, 1992; Kyle, 1985). They recover the expected loss through the bid-ask spread, while price impact records how strongly order flow moves prices. These measures make the trading-cost implications of information asymmetry observable. Costly information prevents markets from becoming fully efficient (Grossman and Stiglitz, 1980a; J. E. Stiglitz, 2000), and liquidity providers revise quotes as the perceived composition of informed and uninformed order flow changes (Hasbrouck, 1991; Easley and O'Hara, 1996). Wider spreads indicate lower market liquidity and can pass through to firms' costs of capital (Tse, 2024; Aliyev, Aquilina, Razayev, and Zhu, 2025; Aquilina, Ibikunle, Rzayev, and Wang, 2025).

The credibility of public signals shapes this information environment. Financial disclosure lowers the cost of acquiring reliable information when investors regard the discloser as credible; weak or non-credible signals raise verification costs and the cost of capital (Diamond and Verrecchia, 1991; Verrecchia, 2001; Healy and Palepu, 2001; Leuz and Verrecchia, 2000; Bushman and Smith, 2001; Goldstein and Yang, 2017). Investors' updates also depend on the perceived bias and authenticity of the source (Dye, 1985; Cologna, Baumberger, Knutti, Oreskes, and Berthold, 2022; Koreh and Mandelkern, 2024). Misinformation can therefore affect liquidity without changing firms' underlying cash flows: it degrades the public signal, increases the cost of separating credible information from false claims, and changes the risk that liquidity providers attach to incoming orders. Recent studies linking digital information and misinformation to liquidity document related effects on market microstructure (Herzing and Mück, 2025; Dees and Ozyurt-Miller, 2026; Mo et al., 2026; Leippold, Wang, and Zhou, 2022).

2.2 Climate Misinformation and Financial Markets

Climate misinformation is a politically produced distortion of that public signal. Organized campaigns by carbon-intensive industries have manufactured uncertainty about climate science and regulation, while partisan media and digital platforms amplify the resulting claims (Oreskes and Conway, 2010; Farrell, 2015; Farrell, 2016; Supran, Rahmstorf, and Oreskes, 2023; Vosoughi, Roy, and Aral, 2018; Acemoglu, Ozdaglar, and ParandehGheibi, 2010). Exposure changes factual beliefs, hardens partisan misperceptions, erodes trust, and resists correction (Flynn et al., 2017; Nyhan and Reifler, 2010; Lewandowsky, Ecker, and Cook, 2017; Bursztyn et al., 2020; Lazer, Baum, Benkler, et al., 2018; Cobb, 2025). Climate misinformation is a political problem because it distorts the evidence citizens use to evaluate policy proposals and hold decision-makers accountable. Organized interests can exploit uncertainty about climate science to weaken support for mitigation and delay regulation (Farrell, McConnell, and Brulle, 2019). Understanding climate change requires interpreting long-term trends and probabilistic evidence; misleading arguments can exploit that difficulty by treating a cold weather event as a refutation of warming or uncertainty as evidence that action is unnecessary. Cook, Ellerton, and Kinkead (2018) show how such claims rely on faulty premises and reasoning. Confusion therefore need not reflect a simple absence of information: apparently plausible arguments and trusted partisan sources can make false claims persuasive. These studies establish the political and behavioral consequences of misinformation, but they do not show how the same content affects the liquidity of firms whose values depend on climate policy.

For businesses, these distortions can cloud expectations about regulation, demand for low-carbon products, and the returns to long-lived investments. That connection is plausible because climate-transition firms are unusually exposed to contested political information. Subsidies, carbon prices, disclosure requirements, and phase-out timetables are set through political institutions and international negotiations. Political and policy uncertainty carries a risk premium and raises volatility (Pastor and Veronesi, 2012; Pástor and Veronesi, 2013; Baker, Bloom, and Davis, 2016), while partisan information and market sentiment can move prices independently of fundamentals (Azzimonti and Fernandes, 2023; Lei and Luo, 2023; De Long, Shleifer, Summers, and Waldmann, 1990). Climate risk, news, and disclosure are likewise reflected in carbon premia, green-bond pricing, divestment, and institutional holdings (Engle et al., 2020; Bolton and Kacperczyk, 2021; Hong, Li, and Xu, 2019; Pankratz et al., 2023; Krueger, Sautner, Tang, and Zhong, 2024; Flammer, 2021; Friede, Busch, and Bassen, 2015; Drahos, 2025). Misinformation differs from ordinary policy uncertainty because it corrupts the signal investors use to evaluate these exposures.

The verification problem is especially salient in fragmented climate governance, where public rules, transnational networks, private standards, taxonomies, and ratings offer competing signals whose authority can be politically contested or vulnerable to capture (Keohane and Victor, 2011; Hale and Held, 2018; Bodansky, Brunnée, and Rajamani, 2016; Rajamani and Brunnée, 2017; Abbott and Snidal, 2000; Bütte and Mattli, 2011). Misinformation then makes genuine sustainability information harder to distinguish from greenwashing (Delmas and Burbano, 2011; Marquis, Toffel, and Zhou, 2016). The unresolved question is whether politically produced climate misinformation changes the trading costs through which markets process that information. The next section derives the competing liquidity predictions.

3 Theory: Climate Misinformation and Market Liquidity

The theoretical construct is information asymmetry: one party holds material information that another lacks (Akerlof, 1970; J. E. Stiglitz, 2000; J. Stiglitz and Kosenko, 2024a; J. Stiglitz and Kosenko, 2024b). The primary dependent variable is not information asymmetry itself, but the firm-day percent quoted bid–ask spread used to observe its liquidity consequences. In the sequential-trade model of Glosten and Milgrom (1985), a market maker posts a bid and an ask without knowing whether the next order comes from an informed or an uninformed trader. The spread compensates for expected losses to informed counterparties: the greater the perceived informed-trading risk, the wider the quote required to break even (Easley and O’Hara, 1992; Hasbrouck, 1991). A wider spread therefore indicates lower market liquidity and greater adverse-selection pressure.

Climate misinformation changes the public signal that investors use to value climate-exposed firms. Subsidy regimes, carbon prices, and regulatory timetables are political outcomes, and misinformation makes claims about those outcomes harder to evaluate (Pastor and Veronesi, 2012; Pástor and Veronesi, 2013; Delmas and Burbano, 2011; Marquis et al., 2016). Investors differ in their ability and willingness to verify these signals. The resulting heterogeneity can affect trading through two competing microstructure channels.

Under the adverse-selection channel, sophisticated investors are better able to identify the credible

climate-pricing signal, while less-informed investors are more likely to trade on false claims. This divergence increases the informational advantage of sophisticated traders. A liquidity provider who expects greater losses when orders originate from better-informed counterparties widens the bid–ask spread and quotes less depth. Per-dollar price impact should move in the same direction because trades reveal more private information. The observable prediction is therefore wider spreads and higher price impact, both indicating lower liquidity (Grossman and Stiglitz, 1980b; Chordia, Roll, and Subrahmanyam, 2000; Han, Hirshleifer, and Walden, 2022; Bordalo, Gennaioli, and Shleifer, 2020). Here, sophistication refers to investors’ capacity to verify information, not to a firm’s climate performance. Larger or wealthier firms may nevertheless have better disclosure, stronger compliance capacity, and a more informed investor base, jointly affecting misinformation exposure and liquidity. Firm fixed effects and the market-capitalization control address stable firm differences and observed size changes, but do not eliminate time-varying differences in disclosure quality or regulatory preparedness. Figure 1 summarizes the proposed adverse-selection pathway.

Under the liquidity-fragility channel, viral misinformation may divert attention from accurate information and attract inexperienced retail traders into climate-exposed stocks (Barber and Odean, 2008; Peng and Xiong, 2006; Sims, 2003). If the uninformed share rises, the market maker’s perceived probability that the next order is informed falls, narrowing the quoted spread. Yet this new order flow is shallow and noisy: each additional dollar traded moves the price by more, increasing price impact and weakening price efficiency. Liquidity fragility is this joint state in which trading appears cheaper at the quoted spread while the market’s capacity to absorb informed flow deteriorates.

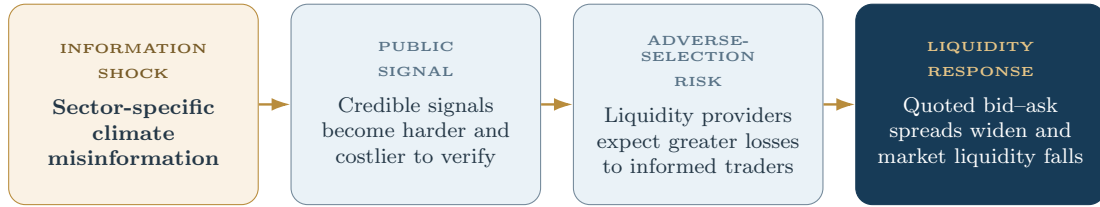


Figure 1: The adverse-selection pathway linking sector-specific climate misinformation to market liquidity.

The first two pathways are distinguished by the sign pattern across the liquidity measures. Adverse selection predicts that quoted spreads and price impact widen together. Liquidity fragility predicts that they move apart: spreads narrow while price impact rises. H1 therefore states that *higher firm-day exposure to sector-specific climate misinformation is associated with narrower percent quoted spreads and higher per-dollar price impact in the same firm-day panel*. The statistical test pairs the spread regression (Model I) with a price-impact specification estimated on the same sample and with the same controls (Section 6.1.1). A widening spread, or the absence of the predicted divergence between the two measures, counts against H1.

Rolling CAPM beta and Fama–French five-factor excess returns provide benchmark outcomes rather than alternative measures of liquidity. A systematic-risk channel should appear in rolling beta (Model II), whereas policy uncertainty, sentiment, or reduced confidence could appear in excess returns (Model III; De Long et al., 1990; Pastor and Veronesi, 2012). Under H1 these loadings should be small because a daily information shock changes the trading environment rather than firms’ return covariances. The climate-finance literature further implies that any benchmark effect should be concentrated where climate news and disagreement are most relevant (Engle et al., 2020; Bolton and Kacperczyk, 2021; Hong et al., 2019; Pankratz et al., 2023). The placebo, regional, and pre/post-2019 checks assess whether the observed pattern is specific to politically contested climate information rather than a general attention effect.

4 Data

The unit of analysis is the firm-day. The estimation sample contains 726,890 firm-day observations on 204 listed climate-transition equities from January 2012 to December 2025. These firms—fossil-fuel incumbents and renewable challengers—are selected from a broader 19,183,738-firm-day universe using nine TRBC energy sub-industries. The financial panel supplies the liquidity outcomes, benchmark outcomes, and controls from Refinitiv Workspace, FRED, the World Bank, and the Kenneth French Data Library. The climate-misinformation corpus supplies the 217,954 unique English-language Facebook posts used to construct the exposure measure; each post’s creation date matches at least one trading day in the

panel. Appendix Tables 19–23 and 24 describe the corpus and its classifications, and Appendix Tables 9–14 summarize the financial panel, its regional and temporal coverage, and the principal variables. All processing and merging is implemented in Python, and the replication code follows the transparency standards of Christensen and Miguel (2018).

4.1 Financial Panel and Outcome Measures

The financial data come from a firm-level panel of 204 listed climate-transition equities sourced from Refinitiv Workspace via the Eikon Python API. The sample is defined by membership in one of nine TRBC energy sub-industries that span the fossil-fuel value chain (Coal, Integrated Oil & Gas, Oil & Gas Drilling, Oil & Gas Exploration and Production, Oil & Gas Refining and Marketing, Oil & Gas Transportation Services, Oil Related Services and Equipment) and the renewable transition (Renewable Energy Equipment & Services, Renewable Fuels). Listing regions cover China (CN), the European Union (EU), Japan (JP), the United Kingdom (UK), and the United States (US). For each security, identified by its RIC code, and each trading day, the panel records open, high, low, and close prices; trading volume and turnover; market capitalization; and dividend yield. Trade-level microstructure variables come from the Refinitiv tick-history database: bid–ask quotes, the volume-weighted average price (VWAP), intraday price moves, buyer- and seller-initiated trade counts, order flow, trade intensity, the bid–ask spread in levels and percent, intraday volatility, and price impact (Karpoff, 1987; Parkinson, 1980). The percent quoted spread—the bid–ask spread divided by the quote midpoint and expressed in percent—is the primary liquidity outcome. Price impact is the companion liquidity outcome required for the joint test of H1. Rolling CAPM beta and Fama–French excess returns provide the benchmark risk and return outcomes. The analysis focuses on climate-transition firms because their cash flows are directly exposed to climate policy and the energy transition that the misinformation corpus contests.

One practical concern is timing. The Shanghai trading day (UTC+8) ends before the New York session (UTC−5) begins, so a same-day merge assigns some misinformation activity that post-dates the Asian close to Asian-market outcomes. The one-day-lag specification in Table 4, estimated on the common firm-day sample, reduces this alignment concern by relating exposure on date t to the next observed trading day. Static firm characteristics—exchange code, TRBC industry classification, country of headquarters, instrument type, and company name—are merged at the security level and are time-invariant within the panel. Region assignments come from RIC suffixes encoding the primary listing exchange, verified against Bloomberg and WRDS definitions for a random subsample of 1,000 firms (99.2% agreement).

Macroeconomic controls come from the FRED API for the United States and the World Bank API elsewhere: GDP growth, unemployment, and CPI inflation for each region (Federal Reserve Bank of St. Louis, 2025; World Bank, 2025). Because many of these series are annual or quarterly, I forward-fill them to the daily panel:

$$M_{r,d} = M_{r,q(d)}, \quad \text{where } q(d) = \max\{q : \text{start}(q) \leq d\}, \quad (1)$$

so that each firm-day observation carries the most recent available value (J. Stock and Watson, 2008). The interpolated macro panel merges with the firm-level data on calendar date and region, with full coverage for all three controls.

Daily Fama–French five-factor returns—market excess return (MktRF), size (SMB), value (HML), profitability (RMW), and investment (CMA)—come from the Kenneth French Data Library.¹ I download six regional factor sets (US, Global, Europe, Japan, Asia Pacific ex-Japan, and North America) and match each firm by listing region and country of headquarters; the match fills 99.0% of the panel. The risk-free rate (R_f) comes from the corresponding regional factor file and is used to compute daily excess returns.

Three variables are constructed from the raw panel. The daily simple return is

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}} \times 100, \quad (2)$$

where $P_{i,t}$ is the closing price of firm i on day t . The daily excess return is $R_{i,t}^e = R_{i,t} - R_{f,t}$, where $R_{f,t}$ is the regional daily risk-free rate from the Fama–French data. The rolling CAPM beta comes from 252-trading-day rolling OLS regressions of firm excess returns on the regional market excess return, requiring a minimum of 126 observations per window:

$$\hat{\beta}_{i,t} = \frac{\text{Cov}(R_{i,s}^e, \text{MktRF}_s)}{\text{Var}(\text{MktRF}_s)}, \quad s \in [t - 251, t]. \quad (3)$$

¹Data available at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

The rolling beta is available for 71.9% of the panel, with a mean of 0.54 and a median of 0.47. This mean sits below the 0.8–1.2 range typical of US equities, for three reasons. The sample is dominated by Chinese (mean $\beta = 0.66$) and Japanese (mean $\beta = 0.42$) firms, where thin trading and non-synchronous price adjustment bias OLS betas downward (Scholes and Williams, 1977; Dimson, 1979). Within the US subsample the mean is 0.49, pulled down by small-cap firms in the lower market-capitalization quintiles. And the rolling window screens out only the most thinly traded firms; applying the Dimson (1979) lead-lag correction would likely raise the mean toward conventional levels. Appendix Table 15 reports mean betas by region and market-capitalization quintile.

The merged climate-transition panel is uniquely indexed by security and date, with no duplicates. China contributes the largest share of firm-day observations, followed by the United States and Japan, and panel composition is stable from 2012 through 2025. Appendix Table 16 reports the composition and coverage of the 204-firm climate-transition sample.

Finally, the climate-transition panel is merged on calendar date and firm sub-industry with the firm-day misinformation-exposure index built from the Meta Content Library, described in Sections 4.2 and 4.3 below. The main regressions use $\text{Exp}_{i,t}^{\text{mean}}$ as the headline regressor of interest, with $\log(1 + \text{Exp}_{i,t}^{\text{sum}})$, $M_{i,t}^{\text{mean}}$, and $R_{i,t}^{\text{mean}}$ as variant specifications, all defined in Section 4.3.

4.2 Climate Misinformation Corpus

The misinformation measure is built from 217,954 unique English-language public Facebook posts—the subset of a 265,652-post corpus extracted from the Meta Content Library (MCL) whose Meta-recorded creation date matches at least one trading day in the climate-transition panel. The corpus was assembled through keyword searches across eight thematic datasets covering climate adaptation, climate politics, global climate justice, resilience, climate activism, and ecological themes; the posts span December 2009 to October 2025 and come from a long-tailed distribution of pages and profiles. Each post carries platform-reported engagement counts—likes, shares, comments, total reactions, and views—plus metadata on content type, surface type, and owner characteristics. Appendix Table 19 reports summary statistics on the corpus as a whole.

The corpus is an English-language, keyword-selected sample of public content, not a representative sample of national Facebook use. The available summaries do not establish the countries of origin of posts or the geographic distribution of their audiences, so differences in platform access, language, and user composition cannot be separated from differences in misinformation production. The exposure measure instead treats Facebook as a potential transnational information environment: claims about energy technologies, carbon policy, or fossil-fuel demand can circulate across borders and inform investors trading firms in the same sub-industry. For example, a claim originating in India could matter for a French-listed company if it reaches that company’s investors or concerns markets or technologies relevant to its cash flows. The link is shared sector relevance and possible cross-border circulation, not the automatic transmission of every foreign post to every firm. The index measures the availability of relevant content, rather than observed investor attention or a country-to-country transmission effect.

The corpus enters this paper at the level of individual posts. Each (post, sub-industry) pair is scored by a single large language model on two continuous 0–100 scales—sector relevance and climate-or-energy misinformation—as described in Section 4.3; the firm-day index used in the empirical models is built from those scores. Three properties of the corpus matter for the analysis. First, the corpus is heavy-tailed in volume by year, with English-language climate-post volume more than tripling after 2019 (Appendix Table 20). Second, the corpus is concentrated in supply: a small set of high-volume pages and profiles produces a large share of total posts (Appendix Table 22), and platform-reported engagement counts are themselves heavy-tailed, with the highest-engagement posts diffusing on the order of three times the mean shares of the median post—a pattern consistent with the finding that misleading content travels disproportionately (Vosoughi et al., 2018). Third, the corpus mixes climate-scientific content, climate-political content, and climate-adjacent content that is on-topic for some sub-industries and entirely off-topic for others; the firm-day construction in Section 4.3 is what makes the sub-industry partition operational.

The classifier uses an open-weight model, and the post-level classifications and firm-day index built from them are released with the replication package. The raw post text falls under the MCL terms of use and is not redistributed. The classifier is assessed with a held-out spot-check (Section 4.3) rather than a hand-coded gold-standard sample (Grimmer, Roberts, and Stewart, 2022). Constructing a gold standard is less straightforward than for a binary label because the task requires two continuous 0–100 judgments—sector relevance and climate-or-energy misinformation—over nuanced content. A hand-coded benchmark

would nevertheless provide a valuable direction for further measurement validation.

4.3 LLM Classification of Misinformation

This section describes the classification framework that turns the corpus into a firm-day measure of misinformation exposure (Wardle and Derakhshan, 2017). The design goal follows directly from the inference problem in Section 5: a regressor that is to vary across firms within a calendar day must be built by asking, for every post, a separate question of every sub-industry the panel contains. That is what the classifier does. It is called once on each of the 1,959,707 (post, sub-industry) pairs—each of the 217,954 paired posts judged against each of the nine TRBC energy sub-industries—and returns, for each pair, how relevant the post is to that sub-industry and how much climate-or-energy misinformation it contains. A market-wide daily index that collapsed a model’s per-post scores into one number would, by construction, assign every firm the same value on a given day and reinstate the firm-invariant problem this paper is designed to avoid; only the per-sub-industry judgment produces the cross-sectional variation the design needs.

A single high-capacity model is used rather than an ensemble. At 1.96 million judgments, the 35-billion-parameter model can distinguish sincere false claims from satire and sector-relevant claims from off-topic content without the prohibitive runtime cost of repeated ensemble inference. Automated detection systems are themselves vulnerable to adversarial manipulation and raise ethical concerns (Qiu, Liu, Zhou, and Huang, 2022; Horne, Nevo, and Smith, 2023), so the design relies on one locally hosted, deterministic classifier whose outputs are reproducible bit-for-bit and whose rubric is fixed in advance and made explicit below.

The model is `Qwen3.6-35B-A3B-FP8`, a mixture-of-experts model with 35 billion total parameters and approximately 3 billion active parameters per token, served in 8-bit floating-point precision on a single NVIDIA Grace-Blackwell GB10 system-on-chip running the vLLM 0.21 inference engine. Inference uses temperature zero and structured-JSON grammar enforcement via the `xgrammar` backend, so each (post, sub-industry) judgment is deterministic, schema-valid, and reproducible (Westwood, Grimmer, and Hall, 2025; Gilardi, Alizadeh, and Kubli, 2023; Santurkar, Durmus, Ladhak, et al., 2023). Each judgment receives the firm’s sub-industry profile (industry name, the set of countries and listing regions of firms in that sub-industry, and a list of example firm names) and the post (text, date, content type, link-attachment caption and description). The rubric instructs the model to score two continuous 0–100 quantities: *sector relevance* $R_{p,s}$, capturing how materially the post’s content relates to firms in sub-industry s , and *climate-or-energy misinformation* $M_{p,s}$, capturing the degree to which the post contains false, misleading, or pseudoscientific claims about climate science or energy.

Two features of the rubric matter. First, misinformation is bounded to climate or energy claims. A post that contains political conspiracy theories unrelated to climate—antisemitic content about EU politicians, anti-vaccination conspiracies, electoral fraud claims—scores $M_{p,s} = 0$ regardless of how false those other claims are; the score is a measure of climate-misinformation exposure, not a general misinformation index. Second, satire and parody are bounded to be not misinformation. A sarcastic Facebook post mocking climate denial that begins “MDPI, the predatory journal of which nothing can be believed, according to Facebook climate experts” is correctly classified at $M_{p,s} = 0$ with the rationale “satirical parody mocking climate denialism, not a sincere assertion of false claims”. These features were added to the rubric after a held-out spot-check on the first 2,000 judgments revealed that a default rubric over-scored political and parody content; the spot-check is reported in Appendix E.

The per-post composite exposure index is

$$\text{Exp}_{p,s} = \frac{R_{p,s} M_{p,s}}{100}, \quad (4)$$

on a 0–100 scale, which is non-zero only when the post is sector-relevant for sub-industry s and contains climate or energy misinformation. The product, rather than either factor alone, is the object of interest because the economic channel requires both: a false climate claim that has nothing to do with coal does not change what a coal trader needs to price, and an accurate, highly relevant post about coal carries no misinformation for the market to misprice. The index isolates the intersection—misinformation that is *about* the firm’s sector. The firm-day regressor used in the empirical models is the mean of this composite across the day’s paired posts, evaluated at the firm’s sub-industry:

$$\text{Exp}_{i,t}^{\text{mean}} = \frac{1}{|P_t|} \sum_{p \in P_t} \text{Exp}_{p,s(i)}, \quad (5)$$

where P_t is the set of climate posts whose Meta-recorded creation date matches calendar date t , and $s(i)$ is the TRBC sub-industry of firm i . Two volume-style variants enter the robustness checks: $\text{Exp}_{i,t}^{\text{sum}} = \sum_p \text{Exp}_{p,s(i)}$ (sensitive to the number of climate posts a firm’s sector sees on a given day) and $\log(1 + \text{Exp}_{i,t}^{\text{sum}})$ (which tames the right skew). The raw components $R_{i,t}^{\text{mean}}$ and $M_{i,t}^{\text{mean}}$ are also persisted, to test whether the composite or one of its constituents carries the signal.

Three properties of the regressor matter for the inference design. First, two firms in different sub-industries on the same date receive different exposures: the model evaluates each post against the specific sub-industry, so a Renewable-Fuels firm and a Coal firm on the same trading day inherit different scores from the same set of posts. Second, two firms in the *same* sub-industry on the same date share a value by construction; the level of aggregation at which the model is asked to reason is the sub-industry, not the firm. Firm-level fixed effects in the empirical models absorb the constant level of exposure that any firm experiences over the sample, leaving within-firm between-day variation as the identifying contrast. Third, the share of firm-days with exactly $\text{Exp}^{\text{mean}} = 0$ is 44% in the climate-transition panel (Appendix Table 25 and Appendix Figure 7)—these are firm-days on which paired climate posts were either off-topic for the firm’s sub-industry or contained no misinformation—so the regressor is sparse, and the volume variants in the robustness section capture how the number of paired posts mediates the response. The classification was carried out over approximately ten days of single-GPU runtime, and 398 of the 1,959,707 (post, sub-industry) judgments returned a parsing error after five retries (0.02%) and are dropped from any downstream firm-day on which they would otherwise have entered the average.

5 Empirical Strategy

The empirical strategy contains two designs that share the firm-day exposure index $\text{Exp}_{i,t}^{\text{mean}}$, firm fixed effects, year-month fixed effects, and two-way clustered inference. Design 1 tests the liquidity mechanisms using the percent quoted spread (Model I) and price impact. Design 2 uses rolling CAPM beta (Model II) and Fama–French excess returns (Model III) as benchmark outcomes for systematic risk and return pricing. Separating the designs keeps the outcome and control set for each test explicit.

5.1 Design 1: Market Liquidity

The baseline spread specification is:

$$\text{SpreadPercent}_{i,t} = \beta_1 \text{Exp}_{i,t}^{\text{mean}} + \gamma' \mathbf{X}_{i,t} + \alpha_i + \delta_t + \varepsilon_{i,t}, \quad (6)$$

where $\mathbf{X}_{i,t}$ contains log trading volume, log market capitalization, intraday volatility, absolute order flow, log turnover, GDP growth, unemployment, and CPI inflation; α_i denotes firm fixed effects; and δ_t denotes year-month fixed effects. All logged variables are natural logarithms. The price-impact test in Section 6.1.1 replaces the outcome with per-dollar price impact while retaining the same estimation sample, controls, fixed effects, and clustering. Adverse selection predicts wider spreads and higher price impact; liquidity fragility predicts narrower spreads and higher price impact. Variant regressors test whether the result is driven by exposure volume, misinformation content, or sector relevance, and are interpreted with the Model I results in Table 1.

5.2 Design 2: Benchmark Risk and Return Outcomes

5.2.1 Model II: Rolling CAPM Beta

The first benchmark asks whether firm-day misinformation exposure is associated with changes in systematic risk. Section 4.1 defines $\hat{\beta}_{i,t}$ from 252-trading-day rolling OLS regressions of firm excess returns on the regional market excess return. Model II uses that rolling beta as the outcome:

$$\hat{\beta}_{i,t} = \lambda \text{Exp}_{i,t}^{\text{mean}} + \gamma' \mathbf{X}_{i,t} + \alpha_i + \delta_t + \varepsilon_{i,t}. \quad (7)$$

The coefficient λ captures the within-firm association between exposure and systematic risk after firm and year-month fixed effects. The control vector adds the percent quoted spread and dividend yield to the common firm-level and macroeconomic controls.

5.2.2 Model III: Augmented Fama–French Five-Factor Model

The baseline Fama–French five-factor model decomposes expected returns as:

$$R_{i,t}^e = \beta_{m,i} \text{MktRF}_t + \beta_{s,i} \text{SMB}_t + \beta_{v,i} \text{HML}_t + \beta_{p,i} \text{RMW}_t + \beta_{c,i} \text{CMA}_t + \varepsilon_{i,t}. \quad (8)$$

The augmented specification introduces the firm-day exposure index alongside the five factors:

$$R_{i,t}^e = \sum_k \beta_k F_{k,t} + \phi \text{Exp}_{i,t}^{\text{mean}} + \gamma' \mathbf{X}_{i,t} + \alpha_i + \delta_t + \varepsilon_{i,t}, \quad (9)$$

where $F_{k,t} \in \{\text{MktRF}, \text{SMB}, \text{HML}, \text{RMW}, \text{CMA}\}$. MktRF denotes the market excess return, SMB (small minus big) the size premium, HML (high minus low) the value premium, RMW (robust minus weak) profitability, and CMA (conservative minus aggressive) investment intensity (Fama and French, 1993a; Fama and French, 2015). Model III uses the same expanded control set as Model II. If misinformation were a priced factor omitted from the standard model, ϕ would be significant. Under H1 the expectation is the opposite: misinformation disrupts the daily trading environment rather than firms’ fundamental risk exposures, so its loading on returns should be muted relative to its association with quoted spreads.

5.3 Identification Strategy

Identification rests on within-firm, within-month variation in the firm-day exposure index. Firm fixed effects α_i absorb all time-invariant heterogeneity (sub-industry, listing exchange, country, instrument type, and the constant level of exposure that any firm in a given sub-industry experiences over the sample). Year-month fixed effects δ_t absorb calendar-time variation in disclosure cycles, macroeconomic regimes, and seasonal patterns in both social-media activity and trading. What remains is the day-to-day movement of $\text{Exp}_{i,t}^{\text{mean}}$ within a calendar month for each firm, conditional on the firm-day controls in $\mathbf{X}_{i,t}$.

The structural feature that distinguishes the firm-day regressor from a market-wide daily index is that two firms in different TRBC sub-industries receive different exposure values on the same calendar day. The coefficient is estimated from residual within-firm changes over time, while the sub-industry construction also provides cross-sectional variation in exposure within a date. Firm fixed effects absorb time-invariant differences across firms, and year-month fixed effects absorb shocks common to a calendar month. Two-way clustering remains necessary to allow residual dependence within firms and across firms observed on the same date (Table 1; Moulton, 1990).

Several checks assess the climate-specific interpretation of H1. If misinformation moves spreads through a channel specific to politically contested climate information, the association should be absent for posts the classifier judges sector-irrelevant. The post-2019 increase in corpus volume and widening fossil-vs.-renewable exposure gap also motivate a subperiod comparison, but do not themselves imply a stronger underlying relationship. The within-classifier placebo of Section 6.1.3 and the subperiod split of Section 6.1.4 provide the primary checks. Appendix Section F reports regional and renewable–fossil decompositions and an exploratory interaction specification. A placebo coefficient of comparable size would count against the climate-specific reading.

The main limitation of the design is that no exogenous shock breaks the joint determination of the regressor and outcome at the firm-day level. The estimates are conditional correlations after firm and year-month fixed effects, not causal effects. The design instead permits comparison of the observed pattern across these checks with the predictions of H1 and the adverse-selection alternative.

6 Results

This section reports the three panel specifications on the 204-firm climate-transition sample. All models are estimated with the `pyfixest` package in Python, with firm (RIC) and year-month fixed effects; continuous controls are winsorized at the 1st and 99th percentiles, and standard errors are clustered two-way by firm and calendar date. Table 1 presents the headline coefficients and variant exposure measures. Sections 6.1–6.3 then report the full estimates, including the price-impact test, one-day lag, within-classifier placebo, and subperiod comparison. Additional renewable–fossil, regional, and interaction checks appear in Appendix Section F.

The headline result is a positive spread coefficient: greater exposure is associated with wider spreads, contrary to H1’s liquidity-fragility prediction. The association is concentrated in 2019–2025 and becomes stronger at a one-day lag. By contrast, the two benchmark outcomes remain weak under two-way clustering. The sections below report the estimates and explain the substantive meaning of each comparison.

Table 1: Climate-Transition Panel, Consolidated Headline Coefficients: Models I–III by IV Variant and Clustering Scheme

Outcome	IV	Firm-clustered SE		Two-way clustered (RIC + Date) SE		N	R^2_{within}
		Coef.	(SE)	Coef.	(SE)		
Spread%	Exp^{mean}	+0.0038**	(0.0016)	+0.0038**	(0.0019)	468,062	0.261
Spread%	$\log(1 + \text{Exp}^{\text{sum}})$	+0.0038***	(0.0011)	+0.0038***	(0.0012)	468,062	0.261
Spread%	M^{mean}	+0.0008**	(0.0004)	+0.0008**	(0.0004)	468,062	0.261
Spread%	R^{mean}	+0.0003	(0.0005)	+0.0003	(0.0005)	468,062	0.261
CAPM β	Exp^{mean}	-0.0007	(0.0009)	-0.0007	(0.0012)	384,017	0.052
CAPM β	$\log(1 + \text{Exp}^{\text{sum}})$	-0.0002	(0.0004)	-0.0002	(0.0005)	384,017	0.052
CAPM β	M^{mean}	-0.0001	(0.0001)	-0.0001	(0.0001)	384,017	0.052
CAPM β	R^{mean}	+0.0008***	(0.0003)	+0.0008**	(0.0003)	384,017	0.052
ExcessReturn	Exp^{mean}	-0.0224***	(0.0053)	-0.0224	(0.0156)	382,890	0.069
ExcessReturn	$\log(1 + \text{Exp}^{\text{sum}})$	-0.0048*	(0.0024)	-0.0048	(0.0089)	382,890	0.069
ExcessReturn	M^{mean}	+0.0033***	(0.0010)	+0.0033	(0.0033)	382,890	0.069
ExcessReturn	R^{mean}	-0.0039***	(0.0009)	-0.0039	(0.0030)	382,890	0.069

Notes: All three models add two-way fixed effects (RIC and year-month). Controls are (Model I) log volume, log market cap, intraday volatility, [order flow], log turnover, CPI inflation, GDP growth, unemployment; (Model II) log volume, log market cap, quoted spread, dividend yield, intraday volatility and the macro controls; (Model III) the Fama–French five factors plus the Model II controls. Numeric controls are winsorized at the 1st and 99th percentiles; the misinformation IVs are not winsorized. Exp^{mean} is the firm-day mean of per-post climate-relevance-weighted misinformation exposure; Exp^{sum} is its sum across the day’s climate posts; M^{mean} is the mean of the per-post misinformation sub-score; R^{mean} is the mean per-post sector-relevance sub-score. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.1 Model I: Misinformation and Quoted Spreads (H1)

Table 2 reports the full readout of Equation 6 on the climate-transition panel. The firm-day mean exposure carries a positive and statistically significant coefficient on the quoted spread under two-way clustering (+0.0038, $p = 0.049$, $N = 468,062$). Under firm-only clustering the coefficient is identical in sign and magnitude and slightly tighter in precision (+0.0038, $p = 0.017$). The log-volume variant is estimated more precisely (+0.0038 on $\log(1 + \text{Exp}^{\text{sum}})$, $p = 0.001$), consistent with a stronger association on higher-volume exposure days while allowing diminishing marginal effects. The pure misinformation sub-score M^{mean} also produces a positive spread coefficient (+0.0008, $p = 0.020$), whereas the sector-relevance sub-score R^{mean} is weakly negative and not statistically significant (-0.0027 , $p = 0.13$ under two-way clustering). Together, these variants indicate that misinformation content, rather than sector relevance alone, is the component most closely associated with the headline result. This suggests that the liquidity cost is linked to the difficulty of assessing misleading climate claims, rather than simply to greater discussion of the energy sector, as the adverse-selection mechanism would imply.

The control coefficients line up with standard microstructure priors. Log turnover is the dominant spread compressor on the climate-transition panel (-0.4501 , $p < 0.001$): more active trading reduces the per-trade adverse-selection component and lowers dealers’ inventory costs (Stoll, 2000; Chordia et al., 2000). Conditional on turnover, log trading volume is positive and insignificant (+0.1628, $p = 0.109$) and log market capitalization is positive and significant (+0.2247, $p = 0.009$); the three activity-and-size measures are strongly collinear within firm, so on a 204-firm panel the compression effect loads onto turnover and the residual partial coefficients on volume and capitalization should not be read structurally. Intraday volatility (+0.0560, $p < 0.001$) and absolute order-flow imbalance (+0.0251, $p < 0.001$) widen spreads: the first captures the inventory-risk premium associated with uncertain prices (Hasbrouck, 1991; Amihud and Mendelson, 1986), and the second proxies the directional information in net order flow (Kyle, 1985; Easley and O’Hara, 1996). Among the macroeconomic controls, CPI inflation enters negatively (-0.0310 , $p = 0.003$), unemployment positively (+0.0201, $p = 0.007$), and GDP growth is negligible (-0.0052 , $p = 0.381$).

The within- R^2 on Model I is 0.26 for the Exp^{mean} specification, so the firm and year-month fixed effects and microstructure controls absorb most of the variation in the quoted spread and the misinformation term enters at the margin. The economic magnitude is modest. A one-standard-deviation increase in $\text{Exp}^{\text{mean}}_{i,t}$ ($\sigma \approx 0.51$ on the climate-transition panel; Appendix Table 25) maps into a quoted-spread response of about 0.0019 percentage points, comparable in size to the order-flow and turnover effects in the same model. Compared with the unconditional mean of the quoted spread on the climate-transition panel ($\overline{\text{Spread\%}} \approx 0.16$), the implied proportional response is $\sim 1.2\%$.

Table 2: Climate-Transition Panel, PanelOLS Full Readout: Model I (Percent Quoted Spread)

Term	Exp ^{mean}			log(1 + Exp ^{sum})		
	Coef.	SE	<i>p</i>	Coef.	SE	<i>p</i>
Exposure term	+0.0038**	0.0019	0.049	+0.0038***	0.0012	0.001
Log Volume	+0.1628	0.1010	0.109	+0.1628	0.1009	0.108
Log Market Cap	+0.2247***	0.0847	0.009	+0.2248***	0.0846	0.009
Intraday Volatility	+0.0560***	0.0052	<0.001	+0.0560***	0.0052	<0.001
Order Flow	+0.0251***	0.0060	<0.001	+0.0251***	0.0060	<0.001
Log Turnover	−0.4501***	0.1026	<0.001	−0.4502***	0.1026	<0.001
CPI Inflation	−0.0310***	0.0102	0.003	−0.0310***	0.0102	0.003
GDP Growth	−0.0052	0.0060	0.381	−0.0053	0.0060	0.377
Unemployment	+0.0201***	0.0074	0.007	+0.0201***	0.0074	0.007
Observations			468,062			468,062
R^2			0.614			0.614
Within R^2			0.261			0.261
Adj. Within R^2			0.261			0.261
Wald F (joint)			21.91			21.77
Firm FE			Yes			Yes
Year-Month FE			Yes			Yes
SE clustering			Firm + Date			Firm + Date

Notes: Dependent variable: percent quoted bid–ask spread, Equation 6, 204-firm climate-transition panel. The two column blocks replace the exposure term with the firm-day mean exposure $\text{Exp}_{i,t}^{\text{mean}}$ and its concave volume variant $\log(1 + \text{Exp}_{i,t}^{\text{sum}})$; control coefficients are nearly identical across the two blocks, as expected by Frisch–Waugh–Lovell when two variants of a sparse firm-day regressor have negligible partial correlation with the controls. Standard errors are CRV1 two-way clustered by firm (RIC) and calendar date. Continuous controls are winsorized at the 1st and 99th percentiles; the exposure regressors are not winsorized. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The sign of the spread coefficient distinguishes the two theoretical mechanisms. Under the theory of Section 3, a quoted spread that widens with sector-specific misinformation exposure is the prediction of the adverse-selection channel: as misinformation circulates around a sub-industry, liquidity providers may perceive greater informed-trading risk and widen their quotes (Glosten and Milgrom, 1985; Easley and O’Hara, 1992). H1, the liquidity-fragility prediction, requires the opposite sign on the spread paired with rising price impact; the price-impact half of that joint test is reported in Section 6.1.1, and the configuration H1 requires does not appear. The spread result therefore rejects H1 for the climate-transition panel under two-way clustering.

6.1.1 Price Impact: Completing the Joint Test of H1

H1 is a joint hypothesis: liquidity fragility predicts that quoted spreads narrow while per-dollar price impact rises, and adverse selection predicts that the two liquidity measures widen together. The spread regression supplies only half of the discriminating evidence. Table 3 therefore re-estimates the Model I specification with per-dollar price impact as the outcome, on the same estimation sample, with the same controls, fixed effects, and two-way clustering.

The contemporaneous price-impact coefficient is statistically indistinguishable from zero (+0.0018, $p = 0.53$ on Exp^{mean} ; +0.0006, $p = 0.67$ on the log-volume variant). At a one-day lag the coefficient grows to +0.0066 but remains short of conventional significance ($p = 0.118$, $N = 374,409$), mirroring the delayed timing of the spread response without matching its precision. The point estimates are positive at every horizon and in every variant—the direction predicted by the adverse-selection channel—but they are too imprecise to provide independent evidence for that mechanism. The joint configuration predicted by liquidity fragility is nevertheless absent because the spread widens significantly rather than narrowing, while price impact is not estimated precisely enough to establish a separate effect. The price-impact controls follow the same general pattern as the spread controls: intraday volatility (+0.1754, $p < 0.001$) and absolute order flow (+0.0351, $p < 0.001$) raise per-dollar price impact while log turnover lowers it (−0.2217, $p = 0.005$). The within- R^2 of 0.391 indicates that these controls absorb a substantial share of within-firm variation.

Table 3: Climate-Transition Panel, PanelOLS Full Readout: Price Impact (Joint Test of H1)

Term	Exp ^{mean}			log(1 + Exp ^{sum})		
	Coef.	SE	<i>p</i>	Coef.	SE	<i>p</i>
Exposure term	+0.0018	0.0028	0.531	+0.0006	0.0015	0.674
Exposure term (lag 1) ^a	+0.0066	0.0042	0.118			
Log Volume	+0.1641**	0.0771	0.035	+0.1641**	0.0771	0.035
Log Market Cap	+0.1013	0.0645	0.118	+0.1013	0.0645	0.118
Intraday Volatility	+0.1754***	0.0075	<0.001	+0.1754***	0.0075	<0.001
Order Flow	+0.0351***	0.0060	<0.001	+0.0351***	0.0060	<0.001
Log Turnover	−0.2217***	0.0785	0.005	−0.2217***	0.0785	0.005
CPI Inflation	−0.0190***	0.0053	<0.001	−0.0190***	0.0053	<0.001
GDP Growth	−0.0106**	0.0050	0.035	−0.0106**	0.0050	0.035
Unemployment	−0.0039	0.0043	0.367	−0.0039	0.0043	0.368
Observations			468,062			468,062
<i>R</i> ²			0.507			0.507
Within <i>R</i> ²			0.391			0.391
Adj. Within <i>R</i> ²			0.391			0.391
Wald <i>F</i> (joint)			232.04			229.56
Firm FE			Yes			Yes
Year-Month FE			Yes			Yes
SE clustering			Firm + Date			Firm + Date

Notes: Dependent variable: per-dollar price impact, estimated with the Model I control set and fixed effects (Equation 6 with PriceImpact as the outcome), 204-firm climate-transition panel. The two column blocks replace the exposure term with the firm-day mean exposure $\text{Exp}_{i,t}^{\text{mean}}$ and its concave volume variant $\log(1 + \text{Exp}_{i,t}^{\text{sum}})$; control coefficients are nearly identical across the two blocks, as expected by Frisch–Waugh–Lovell when two variants of a sparse firm-day regressor have negligible partial correlation with the controls. Standard errors are CRV1 two-way clustered by firm (RIC) and calendar date. Continuous controls are winsorized at the 1st and 99th percentiles; the exposure regressors are not winsorized. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

^a The lag-1 row replaces the contemporaneous exposure with its one-day lag on the restricted sample of 374,409 firm-days for which both exposures are observed, mirroring the lag specification of Table 4.

6.1.2 Lag Specification

Table 4 re-estimates Models I–III with $\text{Exp}_{i,t}^{\text{mean}}$ replaced by its one-day lag, restricting in both columns to firm-days for which both contemporaneous and lagged exposures are observed.

The Model I lagged coefficient is +0.0104 ($p = 0.004$), more than twice the contemporaneous coefficient (+0.0044, $p = 0.033$) on the same restricted sample. The pattern is consistent with delayed quote revision: exposure on date t is reflected more strongly in the quoted spread on date $t+1$. The Model II and Model III lagged coefficients are not distinguishable from zero. The lag specification strengthens temporal ordering and reduces concern that the spread association is entirely contemporaneous, but it does not identify a causal effect because persistent or serially correlated confounders may remain.

6.1.3 Placebo: Sector-Irrelevant Climate Posts

The natural placebo for a firm-day exposure index is the same construction restricted to posts the same classifier judged *not* sector-relevant. Table 5 compares two firm-day means built from disjoint subsets of the classified posts. The “treated” aggregate uses only posts with per-post sector-relevance $R_{p,s} \geq 50$; the “placebo” aggregate uses only posts with $R_{p,s} < 10$. Both aggregates enter the Model I, II, and III specifications of Table 1.

The placebo aggregate is statistically indistinguishable from zero on Model I (−0.018, $p = 0.35$), Model II (−0.003, $p = 0.46$), and Model III (+0.025, $p = 0.61$). The treated aggregate carries a small marginally significant negative coefficient on the spread (−0.0002, $p = 0.08$). Because the relevance threshold of 50 selects a smaller, lower-volume subset, this estimate is not directly comparable to the continuous headline index and does not reproduce its positive association. The null placebo likewise does not reproduce the headline positive spread association, supporting—but not establishing—the climate-specific interpretation.

Table 4: Contemporaneous vs. one-day lag of firm-day misinformation exposure (two-way clustered SE)

Outcome	Contemporaneous IV		One-day lagged IV	
	Coef.	(SE)	Coef.	(SE)
Spread%	+0.0044**	(0.0021)	+0.0104***	(0.0035)
CAPM β	-0.0011	(0.0013)	-0.0016	(0.0017)
ExcessReturn	-0.0156	(0.0154)	-0.0228	(0.0260)
N	374,409 (Model I); slightly fewer for II/III			

Notes: Both columns estimated on the same firm-day sample, restricting to observations with non-missing lagged exposure. Specifications and controls as in Table 1. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Placebo: high-relevance vs. low-relevance climate posts (two-way clustered SE)

IV	Spread%	CAPM β	ExcessReturn
Posts with sector-relevance ≥ 50 (treated)	-0.0002*	-0.0001	-0.0029*
Posts with sector-relevance < 10 (placebo)	-0.0178	-0.0039	+0.1647

Notes: The IV in each row is the firm-day mean exposure computed using only those classified posts whose per-post sector-relevance falls into the indicated band, then merged into the energy panel and used in the Table 1 specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.1.4 Subperiod Stability

Table 6 reports the three models separately on 2012–2018 and 2019–2025 firm-day samples.

Table 6: Subperiod splits: pre-2019 vs. 2019–2025 (two-way clustered SE)

Outcome	2012–2018		2019–2025	
	Coef. (SE)	N	Coef. (SE)	N
Spread%	-0.0005 (0.0019)	193,062	+0.0056** (0.0025)	275,000
CAPM β	+0.0014 (0.0009)	157,871	-0.0077*** (0.0021)	226,146
ExcessReturn	-0.0240 (0.0160)	164,801	-0.0039 (0.0375)	218,088

Notes: The IV is Exp^{mean} . Two-way clustered standard errors on RIC and Date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The Model I coefficient is essentially zero in 2012–2018 (-0.0005 , $p = 0.80$) and significantly positive in 2019–2025 ($+0.0056$, $p = 0.024$). The Model II coefficient is small and insignificant in the early window and significantly negative in the later window (-0.0077 , $p < 0.001$). The Model III coefficient is negative in the early window (-0.024 , $p = 0.14$) and essentially zero in the later window, and is not significant in either sub-window. The post-2019 concentration of the spread effect is the main subperiod result. It coincides with the surge in English-language climate-post volume after 2019, with the share of paired posts the classifier scores at $M \geq 25$ doubling between the two sub-windows, and with the period in which fossil sub-industries’ mean exposure trends sharply upward relative to renewables (Figure 2). Substantively, this divergence suggests that the contested information environment has become more uneven across energy sub-industries: investors valuing fossil-fuel firms encounter a different mix of climate claims from those valuing renewables. This concerns the distribution of measured exposure, not evidence of a larger spread response among fossil firms. More posts may also change measurement precision or corpus composition. A significant coefficient in one period and an insignificant coefficient in another do not establish a significant difference between them; without a formal test of that difference, the estimates do not demonstrate that the underlying spread relationship strengthened after 2019.

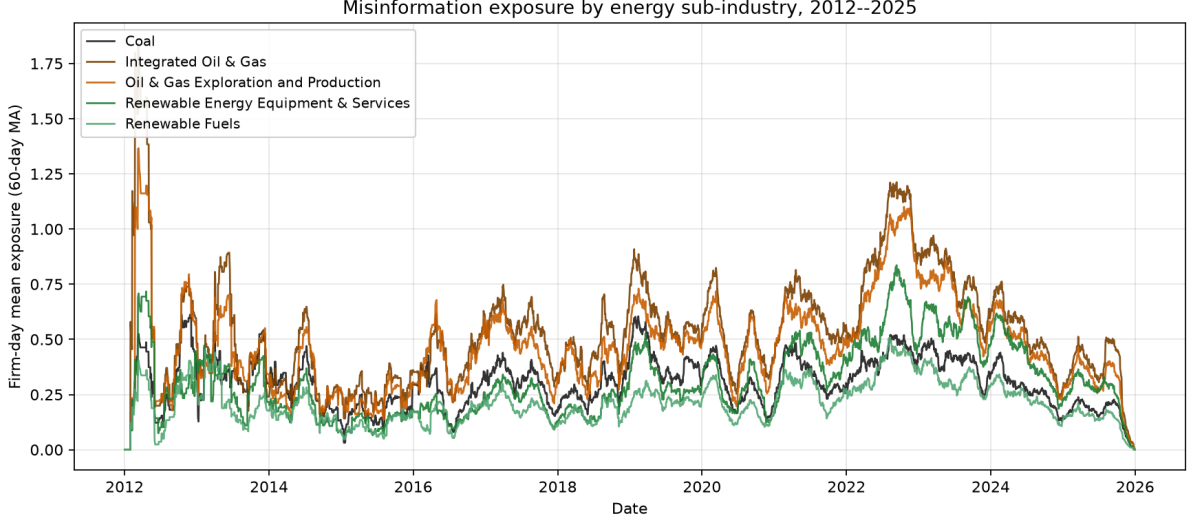


Figure 2: Firm-day mean exposure by climate-transition sub-industry, 2012–2025 (60-day moving average). Fossil sub-industries trend upward after 2019; renewables are flatter. This asymmetry coincides with the post-2019 concentration of the headline spread coefficient.

6.2 Model II: Misinformation and Systematic Risk (CAPM Beta)

Table 7 reports the full readout of Equation 7. The CAPM beta coefficient is not distinguishable from zero in any of the headline specifications: -0.0007 ($p = 0.53$, $N = 384,017$) under Exp^{mean} , -0.0002 ($p = 0.72$) under $\log(1 + \text{Exp}^{\text{sum}})$, and -0.0001 ($p = 0.21$) under M^{mean} . The full climate-transition panel therefore provides no statistically discernible association between day-to-day sector-specific misinformation exposure and rolling CAPM beta. The within- R^2 on Model II is 0.052.

The control coefficients align with standard asset-pricing intuition. Log trading volume is positively associated with beta ($+0.0336$, $p < 0.001$), consistent with actively traded securities tracking aggregate market movements more closely (Chordia et al., 2000; Amihud and Mendelson, 1986). The percent quoted spread enters negatively (-0.0293 , $p = 0.021$), as expected if less liquid stocks adjust more slowly to common shocks (Hasbrouck, 1991; Stoll, 2000). Dividend yield is also negative (-0.0074 , $p = 0.045$), and intraday volatility is positively associated with beta at marginal significance ($+0.0037$, $p = 0.072$). Log market capitalization carries a negative coefficient (-0.0398 , $p = 0.048$), although it is subject to the same collinearity caution as the size-and-activity block in Model I. The three macroeconomic controls all enter negatively (GDP growth -0.0425 , unemployment -0.0297 , CPI inflation -0.0217 , each $p \leq 0.001$), a pattern compatible with counter-cyclical beta dynamics (J. H. Stock and Watson, 2002; Baker et al., 2016). The low within- R^2 reflects the construction of the 252-day rolling beta: it is a slow-moving object, most of its cross-sectional variation is absorbed by firm fixed effects, and day-level regressors explain only the marginal daily perturbation around each firm’s characteristic risk profile.

The post-2019 sub-panel is the exception. The Model II coefficient is significantly negative there (-0.0077 , $p < 0.001$; Section 6.1.4); the same sign appears on the Japanese sub-panel (-0.0035 , $p < 0.001$). One possible explanation is short-lived decoupling from the regional market factor within the rolling estimation window rather than a durable shift in systematic risk. The size of the effect is small ($\Delta\beta \approx -0.004$ for a one-standard-deviation increase in Exp^{mean}), and it is not present in the full-period estimate.

6.3 Model III: Fama–French Excess Returns

Table 8 reports the full readout of Equation 9. The Fama–French excess-return coefficient is negative across IV specifications and shows the largest precision gap between firm-only and two-way clustering. Under firm-only clustering on Exp^{mean} the coefficient is -0.0224 with $p < 0.001$ ($N = 382,890$); under two-way clustering the same coefficient is -0.0224 with $p = 0.151$. The point estimate is unchanged because clustering affects the estimated standard errors rather than the coefficient; the loss of significance indicates a substantial date-level component in the residual dependence. Sector-specific misinformation exposure is associated with a small reduction in same-day excess return ($\Delta R^e \approx -0.011$ percentage

Table 7: Climate-Transition Panel, PanelOLS Full Readout: Model II (CAPM Beta)

Term	Exp ^{mean}			log(1 + Exp ^{sum})		
	Coef.	SE	<i>p</i>	Coef.	SE	<i>p</i>
Exposure term	−0.0007	0.0012	0.525	−0.0002	0.0005	0.716
Log Volume	+0.0336***	0.0056	<0.001	+0.0336***	0.0056	<0.001
Log Market Cap	−0.0398**	0.0200	0.048	−0.0398**	0.0200	0.048
Spread Percent	−0.0293**	0.0126	0.021	−0.0293**	0.0126	0.021
Dividend Yield	−0.0074**	0.0036	0.045	−0.0074**	0.0036	0.045
Intraday Volatility	+0.0037*	0.0020	0.072	+0.0037*	0.0020	0.072
CPI Inflation	−0.0217***	0.0065	0.001	−0.0217***	0.0065	0.001
GDP Growth	−0.0425***	0.0097	<0.001	−0.0425***	0.0097	<0.001
Unemployment	−0.0297***	0.0042	<0.001	−0.0297***	0.0042	<0.001
Observations			384,017			384,017
R^2			0.671			0.671
Within R^2			0.052			0.052
Adj. Within R^2			0.052			0.052
Wald F (joint)			14.26			14.26
Firm FE			Yes			Yes
Year-Month FE			Yes			Yes
SE clustering			Firm + Date			Firm + Date

Notes: Dependent variable: 252-day rolling CAPM beta, Equation 7, 204-firm climate-transition panel. The two column blocks replace the exposure term with the firm-day mean exposure $\text{Exp}_{i,t}^{\text{mean}}$ and its concave volume variant $\log(1 + \text{Exp}_{i,t}^{\text{sum}})$; control coefficients are nearly identical across the two blocks, as expected by Frisch–Waugh–Lovell when two variants of a sparse firm-day regressor have negligible partial correlation with the controls. Standard errors are CRV1 two-way clustered by firm (RIC) and calendar date. Continuous controls are winsorized at the 1st and 99th percentiles; the exposure regressors are not winsorized. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

points for a one-standard-deviation increase in exposure), but the association is not robust to date-level clustering and does not count as a stand-alone finding under the inference standard set in Section 5.

The log-volume and M^{mean} variants of the IV produce smaller and less precise excess-return coefficients (−0.0048, $p = 0.59$ on the log-volume; +0.0033, $p = 0.32$ on M^{mean}). I therefore do not interpret Model III as carrying an independent signal in the firm-day analysis. The role of Model III in the paper is as a falsification benchmark: under H1 a daily information shock should perturb trading, not restructure return covariances, and the absence of a robust excess-return loading is consistent with that prediction even though H1’s spread prediction is itself rejected.

The factor loadings give context for the muted misinformation term. The market loading (+0.6668, $p < 0.001$) says the average climate-transition firm captures about two-thirds of the regional daily market premium. The positive SMB loading (+0.2003, $p < 0.001$) indicates a small-cap tilt, and the large positive HML loading (+0.4195, $p < 0.001$) the pronounced value orientation expected of energy firms with tangible, capital-intensive assets (Fama and French, 1993a; Fama and French, 2015). The profitability factor RMW (+0.0066, $p = 0.848$) and the investment factor CMA (+0.0169, $p = 0.704$) are both null after firm and year-month fixed effects. Among the firm-level controls, the percent quoted spread (+0.1435, $p < 0.001$), log trading volume (+0.1516, $p < 0.001$), log market capitalization (+0.0677, $p = 0.002$), and intraday volatility (+0.0252, $p = 0.021$) are positively associated with excess returns (Amihud and Mendelson, 1986; Parkinson, 1980); dividend yield is negative (−0.0143, $p < 0.001$; Fama and French, 1993b). The three macroeconomic controls are statistically indistinguishable from zero once the five factors and year-month fixed effects are included.

7 Discussion

The strongest evidence concerns quoted spreads. Greater sector-specific misinformation exposure is associated with wider spreads (+0.0038, $p = 0.049$; Table 2), indicating lower market liquidity. The association strengthens at a one-day lag, while the sector-irrelevant placebo is statistically indistinguishable from zero. Price impact is imprecisely estimated but does not display the divergence required by liquidity fragility, so H1 is rejected. The spread pattern is instead consistent with adverse selection:

Table 8: Climate-Transition Panel, PanelOLS Full Readout: Model III (Fama–French Excess Return)

Term	Exp ^{mean}			log(1 + Exp ^{sum})		
	Coef.	SE	<i>p</i>	Coef.	SE	<i>p</i>
Exposure term	−0.0224	0.0156	0.151	−0.0048	0.0089	0.591
FF_MktRF	+0.6668***	0.0439	<0.001	+0.6667***	0.0439	<0.001
FF_SMB	+0.2003***	0.0450	<0.001	+0.2002***	0.0450	<0.001
FF_HML	+0.4195***	0.0481	<0.001	+0.4196***	0.0481	<0.001
FF_RMw	+0.0066	0.0345	0.848	+0.0072	0.0346	0.835
FF_CMA	+0.0169	0.0446	0.704	+0.0171	0.0446	0.702
Log Volume	+0.1516***	0.0159	<0.001	+0.1517***	0.0159	<0.001
Log Market Cap	+0.0677***	0.0215	0.002	+0.0677***	0.0215	0.002
Spread Percent	+0.1435***	0.0279	<0.001	+0.1434***	0.0279	<0.001
Dividend Yield	−0.0143***	0.0029	<0.001	−0.0142***	0.0029	<0.001
Intraday Volatility	+0.0252**	0.0108	0.021	+0.0252**	0.0108	0.021
CPI Inflation	+0.0008	0.0119	0.945	+0.0008	0.0119	0.946
GDP Growth	+0.0077	0.0173	0.656	+0.0077	0.0173	0.658
Unemployment	−0.0068	0.0081	0.403	−0.0068	0.0081	0.400
Observations			382,890			382,890
R^2			0.077			0.077
Within R^2			0.069			0.069
Adj. Within R^2			0.069			0.069
Wald F (joint)			36.49			36.48
Firm FE			Yes			Yes
Year-Month FE			Yes			Yes
SE clustering			Firm + Date			Firm + Date

Notes: Dependent variable: daily Fama–French five-factor excess return, Equation 9, 204-firm climate-transition panel. The two column blocks replace the exposure term with the firm-day mean exposure $\text{Exp}_{i,t}^{\text{mean}}$ and its concave volume variant $\log(1 + \text{Exp}_{i,t}^{\text{sum}})$; control coefficients are nearly identical across the two blocks, as expected by Frisch–Waugh–Lovell when two variants of a sparse firm-day regressor have negligible partial correlation with the controls. Standard errors are CRV1 two-way clustered by firm (RIC) and calendar date. Continuous controls are winsorized at the 1st and 99th percentiles; the exposure regressors are not winsorized. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

when the public signal becomes less reliable, liquidity providers face greater expected losses to better-informed counterparties and widen their quotes (Glosten and Milgrom, 1985; Easley and O’Hara, 1992). Climate claims about carbon restrictions, subsidies, or technology viability may be costly to verify, allowing investors with specialist knowledge to interpret their valuation implications before others. Liquidity providers may then price that informational disadvantage into quotes. This would explain why spreads widen rather than narrow: any influx of uninformed trading may be insufficient to offset the perceived risk of trading against better-informed investors. The stronger lagged association is compatible with verification and quote adjustment taking time, although the imprecise price-impact estimates and absence of direct investor-flow data prevent a definitive separation of the mechanisms.

The benchmark models qualify that interpretation without reproducing the main result. Rolling CAPM beta is essentially unchanged on the full panel (Table 7), indicating that the exposure measure does not generally restructure firms’ covariance with the regional market. Fama–French excess returns are negative under firm-only clustering but lose precision under two-way clustering (Table 8), so Model III does not support an independent return effect. The strongest evidence therefore concerns trading costs rather than systematic risk or return premia.

The political-economy implication is most direct at the sub-industry level. The evidence emerges once the regressor varies within a day across the nine TRBC energy sub-industries—fossil chains versus renewables, integrated production versus refining, and coal versus oil services. A coal-firm trader on a high-exposure day faces a different information environment from a renewable-fuels trader on the same day. A market-wide daily index cannot represent that distinction, which is central to identifying the spread association.

Two checks bear on the climate-specific interpretation. First, when the firm-day index is rebuilt from only those posts the same model judges sector-irrelevant ($R_{p,s} < 10$), the Model I coefficient is statistically indistinguishable from zero (-0.018 , $p = 0.35$). The null placebo does not reproduce the headline positive spread association, supporting—but not establishing—the sector-specific interpretation. Second, the positive spread association is statistically discernible in 2019–2025, alongside greater corpus volume and a wider fossil–renewable exposure gap (Figure 2). Those changes may affect measurement precision and sample composition; they do not establish that the underlying relationship is stronger in that period. The subperiod estimates require a formal coefficient-difference test before they can be interpreted as evidence of a change in the relationship.

The reach-based channel emphasized by Acemoglu et al. (2010) and Vosoughi et al. (2018) has a natural interpretation in the present design. The log-volume variant of the regressor—which is sensitive to the number of sector-relevant climate posts a firm’s sub-industry sees on a given day—has the most precisely estimated Model I coefficient ($p = 0.001$). This pattern may be compositional rather than purely volume-driven because high-volume climate-post days are also high-engagement days. The data do not separate volume from engagement empirically because the classification was performed without engagement weights; doing so would require comparing an engagement-weighted firm-day index with the unweighted measure. The current evidence is consistent with a compositional reading but does not identify it.

For the political economy of climate finance, two implications follow. First, the association between sector-specific climate misinformation and trading costs runs in the direction adverse-selection theory predicts—wider, not narrower, quoted spreads—and is concentrated after 2019, when corpus volume and sub-industry differentiation increase. The regional estimates are directionally suggestive but imprecise. Second, measurement matters as much as model choice. Future work should construct firm-day regressors at the level at which political content varies rather than aggregate across firms whose substantive exposure differs.

8 Conclusions

The results for climate-transition firms show a clear pattern. Firms with greater exposure to climate misinformation have significantly wider quoted bid–ask spreads, indicating lower market liquidity. This relationship remains statistically significant after accounting for firm and time effects and becomes stronger one day after exposure, suggesting that the market’s response is not immediate but strengthens over time. Price impact does not move in the opposite direction, so the divergence between the two liquidity measures required by H1 is absent. The evidence rejects the liquidity-fragility prediction and is more consistent with adverse-selection pressure.

Four conclusions follow. First, evidence that discriminates between H1 and the adverse-selection alternative becomes visible only once the misinformation regressor is constructed with cross-sectional variation

across firms within a calendar day. A market-wide daily index of misinformation diffusion, applied to the same panel of climate-transition firms, returns the opposite sign with no statistical significance, and the two regressors are nearly orthogonal ($r = 0.10$). The political-economy content of climate misinformation is partitioned along the sub-industry axis, not along the calendar-day axis, and the association is most clearly identified when the regressor respects that partition.

Second, the rolling CAPM beta response is small and not robust on the full climate-transition panel. This is consistent with a change in the day’s trading environment rather than a restructuring of rolling return covariances. The exception is the post-2019 sub-panel, where the beta loading is significantly negative; given its small magnitude and absence in the full panel, I do not interpret it as a structural shift in systematic risk.

Third, the Fama–French excess-return response is negative under firm-only clustering but loses precision under two-way clustering. Model III is therefore not a stand-alone finding. The pattern is consistent with H1’s benchmark expectation that a daily information shock should perturb trading rather than restructure return premia, but the present data do not support a robust excess-return interpretation.

Fourth, the post-2019 concentration of the headline spread coefficient is the most pronounced temporal pattern in the analysis. Before 2019 the regressor is not associated with any outcome; after 2019 it is associated with wider spreads and a negative beta coefficient, while the excess-return estimate remains imprecise. The contemporaneous increase in corpus volume and fossil–renewable exposure differences (Figure 2) may change measurement precision and composition, but does not establish a stronger underlying relationship after 2019. The empirical claim is therefore conditional: during 2019–2025, exposure to climate misinformation in the English-language Facebook corpus is associated with trading costs in the direction adverse-selection theory predicts.

For the political economy of climate finance, the daily contest over climate information is reflected in wider quoted spreads among firms whose sub-industries are the subject of the contested content. The estimated magnitude is modest ($\sim 1.2\%$ of the unconditional mean spread for a one-standard-deviation increase in exposure) and is concentrated in the post-2019 period. Classifying misinformation at the sub-industry level preserves cross-sectional variation that a market-wide daily index loses. Future work can move in three directions. An instrumental-variable or event-study design built on exogenous political triggers—a court ruling, a leaked memo, a viral hoax—is needed to move from association to identification. Direct retail-flow data would help distinguish adverse selection from a residual attention channel. The firm-day exposure construction could also be applied to other domains of politically contested knowledge—public health, technology regulation, and financial regulation—to test whether the pattern extends beyond climate (Lins, Servaes, and Tamayo, 2017; Taber and Lodge, 2006).

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Appendix

Section A reports descriptive statistics for the financial panel (Tables 9–14), and Section B reports coverage and composition for the climate-transition equity panel. Section C documents the misinformation corpus (Tables 19–23), including misinformation type and reach in Section C.1. Section D maps the dissemination network, Section E describes and validates the firm-day exposure index, and Section F reports the heterogeneity and interaction checks.

A Financial Panel Descriptive Statistics

The merged firm-level panel comprises 19,183,738 firm-day observations across 5,375 unique equity securities spanning 2012–2025. Table 9 reports summary statistics - including observation counts, fill rates, means, standard deviations, and full distributional percentiles - for nine firm-level trading variables (prices, volume, market capitalization, dividend yield, bid-ask spreads, returns, and CAPM beta), three macroeconomic controls (GDP growth, unemployment, CPI inflation), and the five Fama-French factors (MktRF, SMB, HML, RMW, CMA). The table is presented in landscape orientation to accommodate the eleven-column layout.

The equity universe spans five geographic regions with heterogeneous market structures and data availability. Table 10 reports the number of firm-day observations and unique RICs per region, alongside fill rates for closing prices, CAPM beta, and Fama-French factors. China contributes the largest share of observations (41.2%), followed by the United States (31.7%) and Japan (23.8%), while the UK and EU together account for approximately 3.4% of the sample.

Temporal stability is a desirable property for panel econometric analysis, as changing panel composition can introduce survivorship bias. Table 11 reports annual observation counts, the number of distinct securities, and non-missing rates for key variables across all fourteen sample years. The constant RIC count of 5,375 per year confirms a balanced panel structure, while data fill rates for prices, returns, and betas increase monotonically over time as coverage improves.

Before estimating the main regression specifications, it is useful to examine the bivariate relationships among key continuous variables for evidence of multicollinearity or unexpected sign patterns. Table 12 presents the upper-triangle pairwise correlation matrix for daily returns, CAPM beta, percentage bid-ask spread, volume, market capitalization, and the three primary Fama-French factors. Correlations are generally modest in magnitude, with the largest off-diagonal entries appearing between MktRF and SMB ($r = -0.14$) and between MktRF and HML ($r = -0.15$).

Fama-French factor returns are matched to each firm based on its listing region, with four regional factor sets covering Asia Pacific, the United States, Japan, and Europe. Table 13 reports the number of matched observations, fill rates, and mean daily values for all five factors by region. The US market excess return exhibits the highest mean (0.056% per day), while Japan exhibits the largest standard deviation (1.153%).

The microstructure variables reported in Table 14 include the percent quoted spread and per-dollar price impact used as the primary and companion liquidity outcomes. The table documents observation counts, the number of securities with non-missing data, and distributional summaries for eight trade-level variables: bid-ask spread, percentage spread, order flow, trade intensity, buyer- and seller-initiated trade indicators, volume-weighted average price (VWAP), and price impact. Coverage is near-universal for spread and order-flow variables (5,343 of 5,375 RICs) but lower for trade intensity (2,029 RICs), reflecting cross-exchange differences in tick-level data availability.

Variable	Unit	N	Fill%	Mean	SD	Min	P25	Median	P75	Max
<i>Firm-Level Variables</i>										
Price Close	USD	13,759,934	71.7	388.67	2,128.98	0.01	10.96	30.58	118.26	359,900.00
Volume	shares	18,394,237	95.9	9.03M	60.58M	0.00	0.00	0.823M	5.08M	45,941.76M
Company Market Cap	USD (thousands)	14,263,811	74.4	1,860,901.54	21,101,237.64	14.87	5,032.21	15,080.74	104,672.42	2,464,175,093.59
Dividend yield	%	12,626,836	65.8	2.38	8.49	0.00	0.48	1.72	3.49	20,316.87
BidAskSpread	USD	13,419,146	70.0	4.24	61.02	-268.00	0.01	0.04	0.30	13,550.00
SpreadPercent	%	13,419,145	70.0	0.54	1.95	-200.00	0.07	0.16	0.37	172.14
DailyReturn		13,255,243	69.1	0.09	115.33	-73.29	-1.04	0.00	1.01	419,653.09
CAPM.Beta		13,802,427	71.9	0.54	0.48	-11.74	0.22	0.47	0.82	298.91
TotalReturn		13,735,222	71.6	0.10	113.30	-98.58	-1.04	0.00	1.02	419,653.09
<i>Macroeconomic Controls</i>										
GDP Growth	%	19,183,738	100.0	3.55	2.90	-10.05	1.82	2.79	6.07	8.57
Unemployment	%	19,183,738	100.0	74.87	12.97	53.10	63.52	80.07	82.02	92.19
CPI Inflation	%	19,183,738	100.0	1.89	1.52	-0.23	0.98	1.92	2.62	8.83
<i>Fama-French Five Factors</i>										
FF MktRF	%	18,987,538	99.0	0.03	1.01	-12.01	-0.44	0.05	0.55	10.03
FF SMB	%	18,987,538	99.0	-0.01	0.53	-4.81	-0.30	0.00	0.29	5.71
FF HML	%	18,987,538	99.0	0.01	0.66	-5.03	-0.32	0.00	0.33	6.73
FF RMW	%	18,987,538	99.0	0.01	0.43	-3.76	-0.23	0.00	0.25	4.25
FF CMA	%	18,987,538	99.0	0.00	0.40	-2.92	-0.21	0.00	0.21	2.47

Table 9: Panel A: Summary statistics for key variables in the merged dataset. The sample spans 2012-2025 and covers 5,375 RICs. Fill% indicates the proportion of non-missing observations for each variable. All firm-level variables are at daily frequency.

Region	Obs	Share%	RICs	First	Last	Price%	Beta%	FF%
CN	7,897,640	41.2	2,231	2012-01-02	2025-12-31	62.3	63.0	100.0
US	6,075,274	31.7	1,712	2012-01-02	2025-12-31	71.6	69.7	97.0
JP	4,558,053	23.8	1,253	2012-01-02	2025-12-31	85.2	87.4	99.7
UK	364,446	1.9	100	2012-01-02	2025-12-31	93.4	92.7	100.0
EU	288,325	1.5	79	2012-01-02	2025-12-31	94.2	92.8	100.0
Total	19,183,738	100.0	5,375					

Table 10: Panel B: Coverage by region. Share% is the fraction of total firm-day observations. Price%, Beta%, and FF% indicate the fill rate for Price Close, CAPM Beta, and Fama–French factors, respectively.

Year	Obs	RICs	Price%	Return%	Beta%
2012	1,363,092	5,375	53.6	51.3	25.2
2013	1,356,566	5,375	54.7	52.2	57.3
2014	1,366,584	5,375	55.8	53.3	58.0
2015	1,365,143	5,375	56.9	54.4	59.4
2016	1,367,945	5,375	59.3	56.5	61.2
2017	1,360,941	5,375	64.4	61.8	64.9
2018	1,368,416	5,375	68.1	65.4	70.3
2019	1,373,060	5,375	71.3	68.5	73.4
2020	1,376,595	5,375	75.4	72.7	76.8
2021	1,377,392	5,375	82.4	80.0	83.1
2022	1,373,896	5,375	87.2	84.6	89.3
2023	1,372,026	5,375	90.1	87.4	93.1
2024	1,382,316	5,375	91.8	88.8	96.5
2025	1,379,766	5,375	92.4	89.6	97.7

Table 11: Panel C: Annual coverage. Obs is the number of firm-day observations per year; RICs is the number of distinct securities observed that year. Price%, Return%, and Beta% are non-missing rates for Price Close, DailyReturn, and CAPM Beta.

	Return	Beta	Spread%	Volume	MktCap	MktRF	SMB	HML
Return	1.00	0.17	0.02	0.00	0.03	0.00	-0.00	-0.00
Beta		1.00	-0.23	-0.01	-0.05	-0.00	-0.00	-0.00
Spread%			1.00	-0.02	0.01	0.00	-0.00	-0.01
Volume				1.00	0.07	-0.00	0.00	0.00
MktCap					1.00	0.00	-0.00	-0.00
MktRF						1.00	-0.14	-0.15
SMB							1.00	0.05
HML								1.00

Table 12: Panel D: Pairwise correlations among key variables. Empty cells below the diagonal are suppressed for readability.

FF Region	Obs	Fill%	MktRF	SD(MktRF)	SMB	HML	RMW	CMA
AsiaPac	7,897,640	100.0	0.019	0.856	-0.007	0.021	0.018	0.006
US	6,075,274	97.0	0.056	1.084	-0.009	-0.003	0.006	-0.003
Japan	4,558,053	99.7	0.033	1.153	0.002	0.012	0.000	0.005
Europe	652,771	100.0	0.033	1.028	0.000	0.009	0.006	0.001

Table 13: Panel E: Fama-French five-factor summary by region mapping. Mean daily factor returns are reported. Fill% indicates the proportion of firm-day observations matched to factors.

Variable	N	RICs	Mean	SD	Median
BidAskSpread	13,419,146	5,343	4.24	61.02	0.04
SpreadPercent	13,419,145	5,343	0.54	1.95	0.16
OrderFlow	13,268,923	5,343	0.08	193.08	0.00
TradeIntensity	5,146,749	2,029	4,750.04	12,478.89	2,198.00
BuyerInitiated	13,837,387	5,343	0.48	0.50	0.00
SellerInitiated	13,837,387	5,343	0.47	0.50	0.00
VWAP	13,584,674	5,343	392.92	2,115.66	30.53
PriceImpact	13,268,922	5,343	0.68	47.95	0.37

Table 14: Panel F: Microstructure-variable coverage. The percent quoted spread and price impact are the liquidity outcomes used in the main analysis. N is the number of non-missing observations; RICs is the number of distinct securities with at least one non-missing value.

Table 15: Mean CAPM Beta by Region and Market-Capitalisation Quintile

Region	Mean β	Median β	Q1 (Small)	Q2	Q3	Q5 (Large)
CN	0.655	0.594	0.643	0.673	0.667	0.600
US	0.488	0.388	0.806	0.837	0.823	0.166
JP	0.415	0.330	0.263	0.283	0.303	0.825
UK	0.747	0.714	0.731	0.785	0.782	0.570
EU	0.905	0.877	0.816	0.967	0.897	0.995

Notes: Mean rolling CAPM beta computed from 252-day OLS windows requiring minimum 126 observations. Quintiles defined by Company Market Cap within each region-day. The low overall mean (0.54) is driven by Japanese firms (0.42) and the preponderance of small-cap US firms in the sample. Non-synchronous trading bias (Scholes and Williams, 1977) likely contributes to downward bias in thinly traded securities.

B Climate-Transition Panel Descriptive Statistics

This subsection reports descriptive statistics for the unique climate-transition firm panel used in the subsample regressions. The objective is to document coverage, composition, and distributional properties before estimation. Across all three tables below, the sample is defined by renewable and fossil-fuel TRBC mappings with utilities excluded, consistent with the main-text climate-transition specifications.

Table 16: Climate-Transition Panel Coverage Summary

Metric	Value
Firm-day observations	726,890
Unique firms (RICs)	204
Start date	2012-01-02
End date	2025-12-31
Unique year-month bins	168
Median observations per firm	3,588

Notes: Coverage statistics for the climate-transition panel used in the fixed-effects regressions.

Table 17: Climate-Transition Panel Composition by Region (Unique Firms)

Region	Unique firms	Share (%)
US	102	50.00
CN	83	40.69
JP	15	7.35
EU	2	0.98
UK	2	0.98

Notes: Shares are calculated over the 204 unique firms in the climate-transition sample.

Table 18: Climate-Transition Panel Key Variable Descriptive Statistics

Variable	N	Mean	Std. Dev.	P25	Median	P75
SpreadPercent	504,043	0.6147	1.6705	0.0607	0.1590	0.4090
CAPM_Beta	525,697	0.5679	0.4984	0.2039	0.5168	0.8864
ExcessReturn	506,612	0.0328	2.6949	-1.2207	-0.0100	1.1272
log_volume	698,179	11.3694	6.8472	5.7071	14.5147	16.3631
log_mktcap	559,387	25.0607	2.7631	22.9046	24.3402	27.1216
IntradayVolatility	496,614	3.4611	2.5748	1.7758	2.7528	4.3222
DivYield	464,457	2.9153	3.6049	0.2509	1.8536	4.1325
cpi_inflation	726,890	2.1371	1.5976	1.2616	2.0000	2.6195
gdp_growth	726,890	3.8414	2.6520	2.3402	2.8876	6.0685
unemployment	726,890	73.1123	11.1097	63.5234	80.0325	80.4977

Notes: Statistics correspond to the estimation sample after standard pre-processing used in the econometric pipeline (including transformations and winsorisation for relevant continuous variables).

The descriptive profile in Tables 16–18 motivates the energy-subsample strategy. The panel is sufficiently long and dense for two-way FE estimation, yet composition is concentrated in US and Chinese firms, making firm-level fixed effects and clustered inference essential.

C Descriptive Statistics and Supporting Tables

The corpus comprises 265,652 public Facebook posts with platform-reported engagement data. Table 19 provides distributional summaries for all engagement metrics - likes, shares, comments, total reactions

(including emoji-specific breakdowns for angry, love, haha, sad, wow, and care reactions), and view counts - as well as text-level characteristics (character length and word count). The large gap between mean and median values for most engagement variables reflects the heavy right skew characteristic of social media diffusion, where a small number of viral posts account for a disproportionate share of total engagement.

Table 19: Social Media Post Summary Statistics

Variable	N	Mean	Median	Std.Dev.	Min	Max
Likes	262,461	140.8	10.00	2,015.6	0.000	525,472.0
Shares	265,651	44.90	2.00	682.3	0.000	81,948.0
Comments	265,651	22.72	0.000	217.6	0.000	39,790.0
Total Reactions	262,492	175.9	13.00	2,278.2	0.000	531,508.0
Views	220,511	10,372.8	1,625.0	91,127.8	101.0	15,900,642
Angry Reactions	262,461	7.69	0.000	225.5	0.000	57,470.0
Love Reactions	262,461	15.38	0.000	267.5	0.000	46,490.0
Haha Reactions	262,461	3.84	0.000	89.18	0.000	31,575.0
Sad Reactions	262,461	3.52	0.000	162.2	0.000	44,420.0
Wow Reactions	262,461	1.66	0.000	31.79	0.000	7,074.0
Care Reactions	262,461	0.000	0.000	0.000	0.000	0.000
Text Length (chars)	265,652	3,599.1	1,337.0	6,828.6	2.00	168,477.0
Word Count	265,652	553.9	197.0	1,062.6	1.00	26,148.0

Notes: Summary statistics for 265,652 social media posts from the Meta Content Library (MCL). Engagement metrics reflect platform-reported counts at the time of data collection.

The 217,954 unique posts paired by calendar date to the climate-transition panel span 2012 through 2025. Table 20 reports their annual distribution, with the misinformation classification produced by the Qwen3.6-35B-A3B model (Section 4.3): a post is counted as misinformation if its continuous misinformation score reaches the rubric’s “clearly false or pseudoscientific climate/energy claims” threshold of 50 on the 0–100 scale. Post volume rises sharply after 2019, coinciding with the intensification of global climate activism and the digital migration associated with the COVID-19 pandemic, while Qwen-classified misinformation prevalence fluctuates between 1.7% and 5.0% across years. The within-year movement reflects climate-political intensity rather than raw volume: prevalence peaks in the contested-policy years 2012 (5.0%), 2017 (3.0%), 2019 (3.2%), and 2022 (3.0%), and falls to its sample minimum in the highest-volume years 2024–2025 ($\approx 1.7\%$), where general climate content grows faster than misinformation.

Table 20: Annual Distribution of Posts and Qwen-Classified Misinformation Prevalence

Year	Total Posts	Misinfo. Posts	Prevalence (%)	Mean Likes	Mean Shares
2012	1,276	64	5.02	86.2	38.96
2013	2,649	82	3.10	87.5	39.97
2014	3,885	88	2.27	87.8	33.56
2015	5,106	95	1.86	115.5	47.93
2016	9,954	272	2.73	138.2	82.97
2017	13,845	413	2.98	93.7	58.13
2018	11,429	268	2.34	66.2	45.01
2019	17,160	542	3.16	90.6	47.83
2020	17,118	427	2.49	128.0	61.19
2021	23,038	469	2.04	112.9	38.67
2022	24,102	710	2.95	86.9	22.68
2023	25,846	731	2.83	76.7	21.54
2024	30,404	538	1.77	88.3	26.95
2025	32,142	552	1.72	344.6	62.91
Total	217,954	5,251	2.41	132.4	43.03

Notes: Annual counts for the 217,954 unique posts paired by date to the climate-transition panel. “Misinfo. Posts” counts posts whose Qwen3.6-35B-A3B misinformation score reaches the threshold of 50; “Prevalence” is that count as a share of the year’s posts.

The Meta Content Library organizes posts into eight keyword-based thematic datasets covering different dimensions of climate discourse. Table 21 breaks down post coverage, Qwen-classified misinformation counts, engagement, and unique owner counts by source dataset. The pattern is intuitive given how the model reasons about content: the explicitly political “combined” multi-keyword set (6.4% prevalence) and the climate-politics set (1.9%) concentrate the misinformation, because that is where contested claims about climate science and policy are made, whereas the adaptation, resilience, and ecological datasets—which collect operational and scientific content—sit at or below 0.2% prevalence. The classifier is not flagging climate *topicality*; it is flagging the false or misleading *claims* that cluster in the political corners of the corpus.

Table 21: Post Coverage by MCL Source Dataset (Qwen-Classified Misinformation)

Source Dataset	Posts	Misinfo.	Prev. (%)	Mean Likes	Mean Shares	Owners
Combined	65,688	4,180	6.4	144.8	69.73	25,171
Climate Change Adaptation Al..	61,079	126	0.2	48.10	11.12	17,931
Climate Change Politics All ..	32,930	618	1.9	153.2	69.05	14,527
Climate Change Global Justic..	14,406	134	0.9	120.4	38.57	7,369
Climate Change Global Resili..	13,448	18	0.1	68.60	11.55	7,019
Climate Change Activism All ..	11,222	149	1.3	639.8	62.98	6,120
Ecological Resilience All Time	9,835	8	0.1	77.90	16.27	5,710
Ecological Justice All Time	9,346	18	0.2	77.30	28.76	4,984

Notes: The 217,954 paired posts drawn from eight MCL keyword-based search datasets. “Combined” denotes the multi-keyword merged set. “Misinfo.” counts posts with Qwen3.6-35B-A3B misinformation score ≥ 50 ; “Owners” is the count of distinct posting accounts per dataset.

Understanding who produces misinformation requires examining the distribution of posting activity across the 55,699 unique post owners in the paired corpus. Table 22 groups owners into five activity brackets and reports each bracket’s contribution to total volume and Qwen-classified misinformation output. The distribution is heavily right-skewed: over half of all owners contribute only a single post, while 124 heavy posters (more than 100 posts each) account for roughly 12% of total volume. The Qwen misinformation rate is broadly stable across brackets (1.3%–2.6%), which carries a substantive point—misinformation in this corpus is not the product of a small set of hyperactive accounts but is distributed across the full activity range, including the long tail of one-time posters. The dissemination problem is therefore one of breadth rather than concentration in supply, even if reach (Section C.1) is far more skewed than production.

Table 22: Post Owner Activity and Qwen-Classified Misinformation Concentration

Activity Bracket	Owners	Total Posts	% of Corpus	Misinfo. Posts	Misinfo. Rate (%)
1 post	30,528	30,528	14.0	669	2.19
2–10 posts	21,627	78,690	36.1	2,000	2.54
11–50 posts	3,152	63,464	29.1	1,639	2.58
51–100 posts	268	18,280	8.4	233	1.27
>100 posts	124	26,990	12.4	710	2.63
Total	55,699	217,952	100.0	5,251	2.41

Notes: Owners of the 217,954 paired posts (two posts have no owner identifier) grouped by posting frequency. Misinformation rate is the share with Qwen3.6-35B-A3B misinformation score ≥ 50 .

The corpus spans multiple content formats and platform surface types, which may shape how misinformation circulates. Table 23 reports the distribution of posts across three metadata dimensions: content type (photos, links, albums, status updates, videos, reshares, and others), surface type (page, profile, group, or event), and owner type (page or profile). Photos are the most common content format at 35.4%, and pages dominate both surface type (90.1%) and owner type (91.7%), reflecting the institutional character of climate-related Facebook activity.

Table 23: Post Metadata: Content Type, Surface Type, and Owner Type

Content	N	%	Surface	N	%	Owner	N	%
photos	94,095	35.4	page	239,340	90.1	Page	243,708	91.7
links	73,672	27.7	profile	19,724	7.4	Profile	21,944	8.3
albums	46,457	17.5	group	6,236	2.3			
status	26,390	9.9	event	352	0.1			
videos	16,524	6.2						
reshare	5,444	2.0						
others	15	0.0						

Notes: Distribution by content type, surface type, and owner type. Pages dominate (91.7%).

C.1 Misinformation Type and Reach in the Qwen Classification

The Qwen3.6-35B-A3B classifier returns more than a binary label. Alongside the continuous 0–100 misinformation score, it assigns each flagged post one of six type labels—denial, conspiracy, misleading framing, pseudoscience, greenwashing, or other—which makes the *kind* of climate misinformation in the corpus observable, not just its quantity. Table 24 reports the composition. Outright *denial* of climate science is the single largest category (49.8% of flagged posts), followed by *conspiracy* framings (30.8%) and *misleading framing* of otherwise-accurate facts (17.8%); pseudoscience and greenwashing are rare. The intuition the typology captures is that the misinformation reaching these markets is predominantly the hard-denial and conspiracy content that organized political campaigns produce, not the softer greenwashing that corporate communications produce—a distinction a binary classifier cannot draw.

Table 24: Composition of Qwen-Classified Climate Misinformation by Type

Type	Posts	% of Misinfo.	% of Corpus	Mean Likes	Mean Shares
Denial	2,615	49.8	1.200	226.1	129.6
Conspiracy	1,616	30.8	0.741	122.2	103.9
Misleading framing	937	17.8	0.430	181.2	115.2
Pseudoscience	81	1.5	0.037	290.1	42.9
Greenwashing	2	0.0	0.001	19.5	11.0
All misinformation	5,251	100.0	2.410	187.2	117.7

Notes: Posts with Qwen3.6-35B-A3B misinformation score ≥ 50 , by the model’s modal non-null type label. “% of Corpus” is the share of all 217,954 paired posts. Engagement is platform-reported at collection.

Two further facts about the Qwen-classified corpus motivate the engagement-weighting logic used in the main text. First, misinformation travels further than other climate content: the 5,251 flagged posts average 187 likes and 118 shares against 131 likes and 41 shares for the non-flagged remainder, and their median share count is 11 against 2—a roughly fivefold gap at the median that is consistent with the broader finding that false content diffuses disproportionately (Vosoughi et al., 2018). Second, the within-type engagement ordering is itself informative: denial and conspiracy posts, the categories produced by politically motivated actors, are shared two-to-three times as often as the corpus average, whereas pseudoscience—which is read and liked but less often forwarded—has the highest mean likes but the lowest mean shares of any misinformation type. Reach, not production, is where the asymmetry lives.

Figure 3 plots monthly Qwen misinformation prevalence over the sample. Prevalence is visibly episodic rather than trending: it spikes around contested-policy episodes and recedes between them, and the post-2019 surge in raw post volume (vertical line) dilutes prevalence even as the absolute count of misinformation posts rises. Figure 4 decomposes the annual flagged count by type, showing that denial and conspiracy content drive both the level and the post-2019 growth of misinformation in the corpus, while misleading framing grows more slowly and pseudoscience stays negligible.

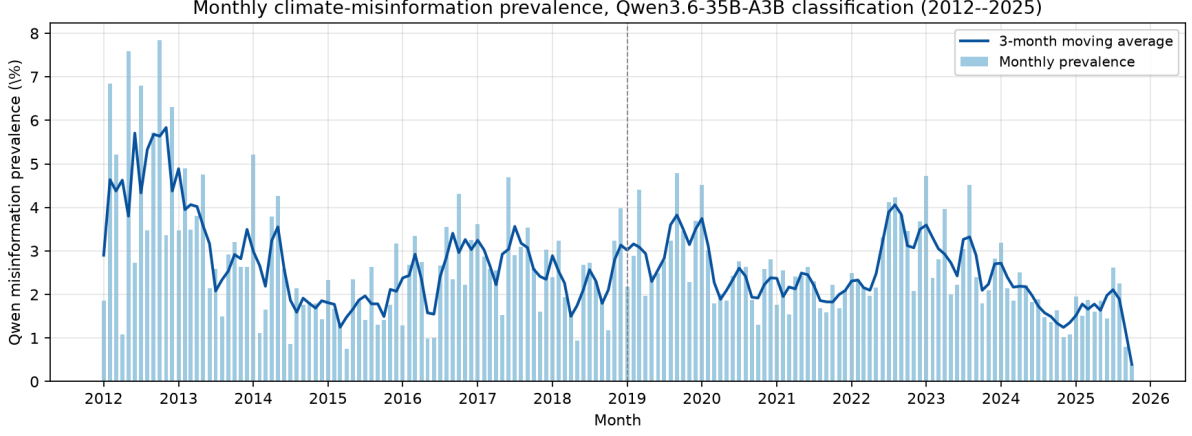


Figure 3: Monthly climate-misinformation prevalence in the Qwen3.6-35B-A3B classification, 2012–2025. Bars are monthly prevalence (flagged posts as a share of that month’s paired posts); the solid line is a three-month moving average. The dashed line marks January 2019.

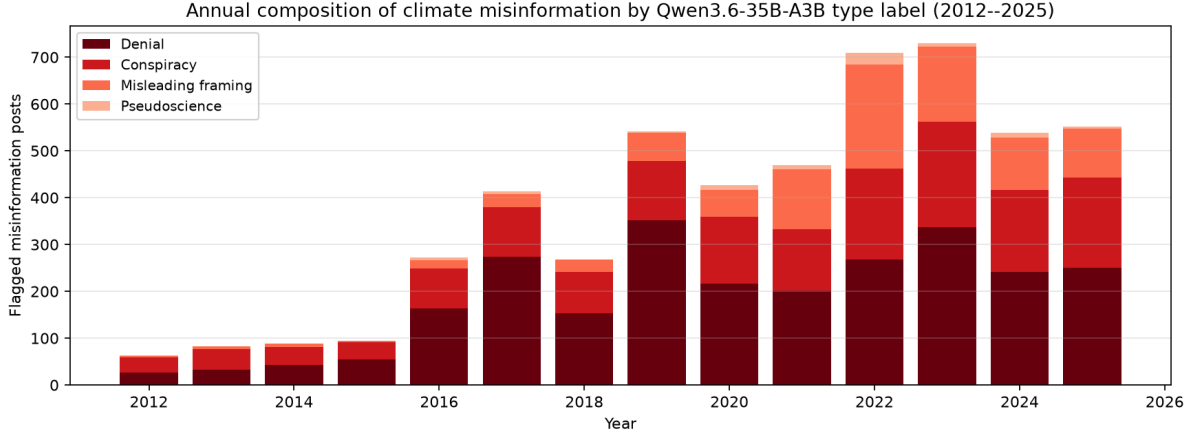


Figure 4: Annual composition of Qwen-classified climate misinformation by type, 2012–2025. Denial and conspiracy framings account for the bulk of flagged content and for its post-2019 growth.

D Network Structure of Misinformation Dissemination

I supplement the post-level corpus with an owner-based network constructed on the 150 most active and engaged post owners (pages and profiles), selected via a composite ranking weighting total engagement (40%), misinformation post count (40%), and overall posting activity (20%). Two owners are connected if their monthly posting distributions correlate at $\rho \geq 0.3$ (Pearson). The largest connected component contains 142 nodes and 1,004 edges, with five Louvain communities (Blondel, Guillaume, Lambiotte, and Lefebvre, 2008) of community misinformation rates varying from 4.2% to 21.7%. The threshold captures accounts that post more during the same months, which may reflect coordinated dissemination, shared thematic interests, or common responses to COP meetings, extreme weather, or elections; the network structure should therefore be read as thematic co-occurrence rather than necessarily coordinated inauthentic behaviour (Walter, Cohen, Holbert, and Morag, 2020; Williams, McMurray, Kurz, and Lambert, 2015).

Three centrality measures bear on the empirical specification that follows. Eigenvector centrality is negatively correlated with the per-owner misinformation rate ($r = -0.296$, $p = 0.0004$), and degree centrality is positively correlated with the absolute misinformation count ($r = 0.218$, $p = 0.009$); betweenness centrality shows no significant association ($r = -0.068$, $p = 0.42$). The structurally central nodes disseminate proportionally lower rates of misleading content, while peripheral high-degree nodes generate larger absolute volumes. This pattern weakens, rather than supports, the prior assumption that engagement-weighting captures “reach-amplified misinformation”: if reach were dominated by misinformation-heavy

actors, eigenvector centrality should covary positively, not negatively, with misinformation rate. Figures 5 and 6 display the network and centrality scatterplots respectively.

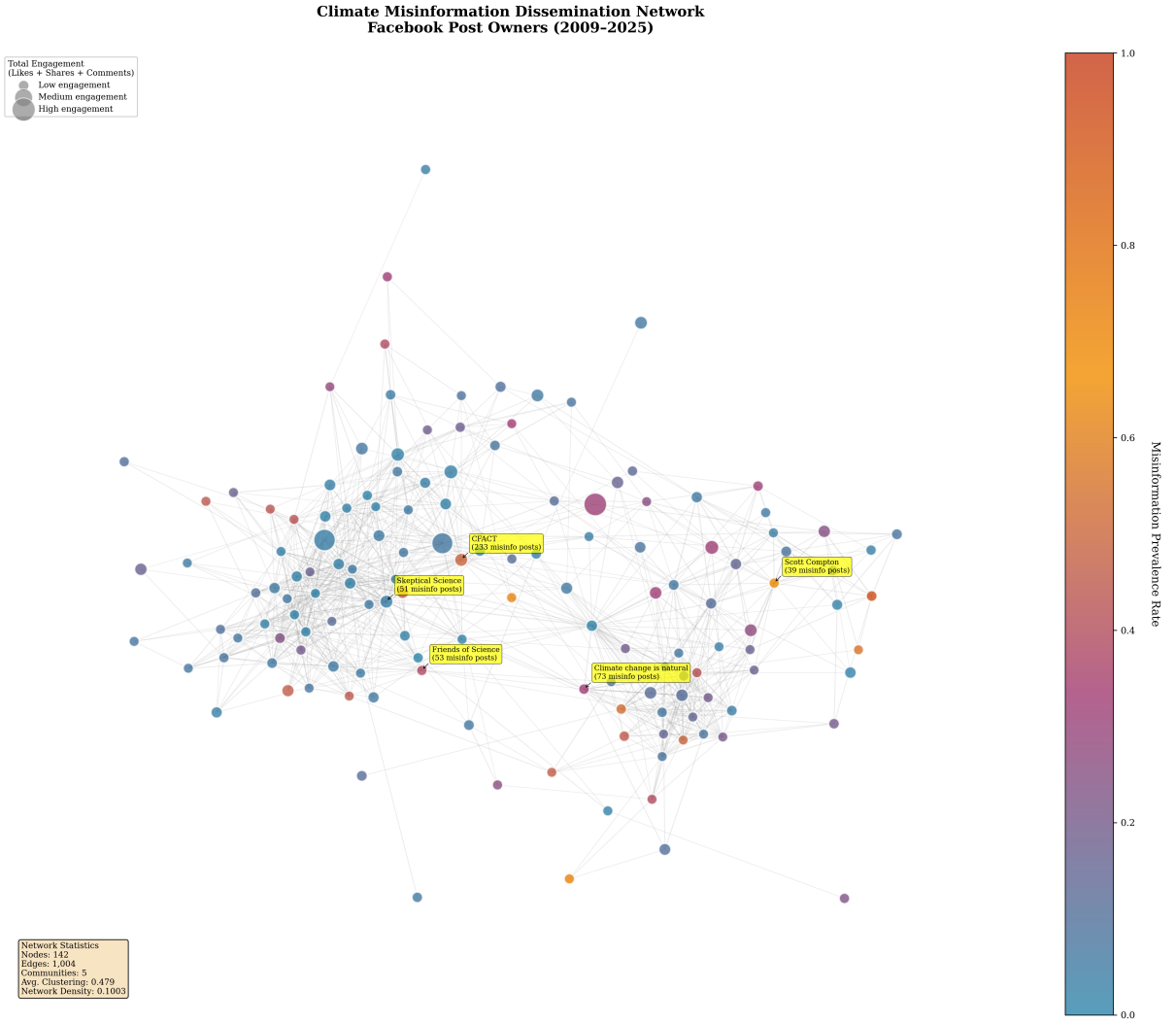


Figure 5: Climate Misinformation Dissemination Network

Notes: Network graph of 142 Facebook post owners (pages/profiles) constructed from temporal posting correlations (Pearson $\rho \geq 0.3$). Node size is proportional to total engagement (likes + shares + comments). Node color indicates misinformation prevalence rate (blue = low, red = high). Edge thickness reflects correlation strength. Top 5 misinformation spreaders are labeled. Force-directed layout (Fruchterman-Reingold). Network statistics: 5 communities (Louvain), average clustering = 0.48, density = 0.10. Data source: Meta Content Library, Dec 2009-Oct 2025.

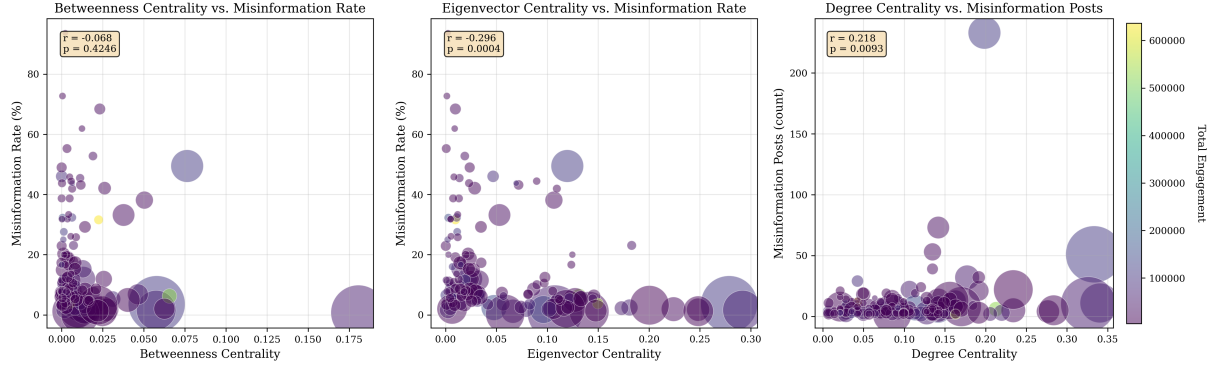


Figure 6: Network Centrality and Misinformation Prevalence

Notes: Scatterplots of three centrality measures against misinformation prevalence and count. **Left panel:** Betweenness centrality vs. misinformation rate ($r = -0.068$, $p = 0.42$). **Center panel:** Eigenvector centrality vs. misinformation rate ($r = -0.296$, $p = 0.0004$). **Right panel:** Degree centrality vs. misinformation post count ($r = 0.218$, $p = 0.009$). Node size represents total posts. Color indicates total engagement. Eigenvector centrality exhibits a statistically significant negative association with misinformation rate, suggesting influential nodes spread less misinformation.

E Firm-Day Exposure: Descriptive Statistics and Validation

This appendix reports the descriptive statistics of the firm-day misinformation-exposure index introduced in Section 4.3 and the held-out spot-check used to validate the classifier’s handling of satire and politically loaded but non-climate content.

E.1 Descriptive Statistics

Table 25 summarises the firm-day index and its constituent scores across the 714,227 firm-days in the climate-transition panel with at least one paired post. The index $\text{Exp}_{i,t}^{\text{mean}}$ has a mean of 0.34 on a 0–100 scale, a median of 0.09, a 99th percentile of 2.73, and a share of exactly-zero firm-days of 44%. The sum variant $\text{Exp}_{i,t}^{\text{sum}}$ has mean 21.85 and 99th percentile 171.50; the log-transformed version $\log(1 + \text{Exp}_{i,t}^{\text{sum}})$ is approximately normal in shape with mean 1.80 and 99th percentile 5.15. The misinformation sub-score $M_{i,t}^{\text{mean}}$ has mean 1.73, p99 15.60, and a share of exactly-zero firm-days of 41%, reflecting the rubric’s restriction of misinformation scoring to climate-or-energy claims. The relevance sub-score $R_{i,t}^{\text{mean}}$ has mean 12.59 and p99 27.17, with only 2% of firm-days exactly at zero, indicating that climate posts paired by date are typically at least weakly relevant to the firm’s sub-industry.

Table 25: Descriptive statistics of the firm-day misinformation-exposure IV (climate-transition panel, 714,227 firm-days with ≥ 1 climate post)

Variable	Mean	SD	p_{50}	p_{90}	p_{99}	Share zero
# climate posts paired to the firm-day	59.900	49.449	52.000	127.000	208.000	0.017
Mean per-post sector relevance R^{mean}	12.594	5.106	12.231	18.524	27.173	0.022
Mean per-post climate misinformation M^{mean}	1.731	3.232	0.781	4.268	15.600	0.408
Firm-day mean exposure Exp^{mean}	0.338	0.715	0.090	0.971	2.732	0.442
Firm-day max exposure Exp^{max}	13.526	20.236	7.000	50.700	68.640	0.442
Firm-day sum exposure Exp^{sum}	21.846	37.661	7.500	71.250	171.500	0.442
Share of posts with $M \geq 25$	0.020	0.037	0.009	0.050	0.167	0.464
$\log(1 + \text{Exp}^{\text{sum}})$	1.795	1.758	2.140	4.280	5.150	0.442

Notes: Sample is firm-days in the energy panel for which at least one English-language Facebook climate post on the same calendar date was scored against the firm’s TRBC industry. “Share zero” is the fraction of firm-days for which the variable evaluates to exactly 0 (typically because no posts in the day’s classification batch were judged sector-relevant or misinformation-laden).

Figure 7 plots the distribution of the headline index and its log-volume variant on non-zero firm-days. The mean variant has a long right tail truncated by the bound of the 0–100 scale; the log-volume variant is more symmetric, justifying its use in the variant Model I specification reported in Table 1.

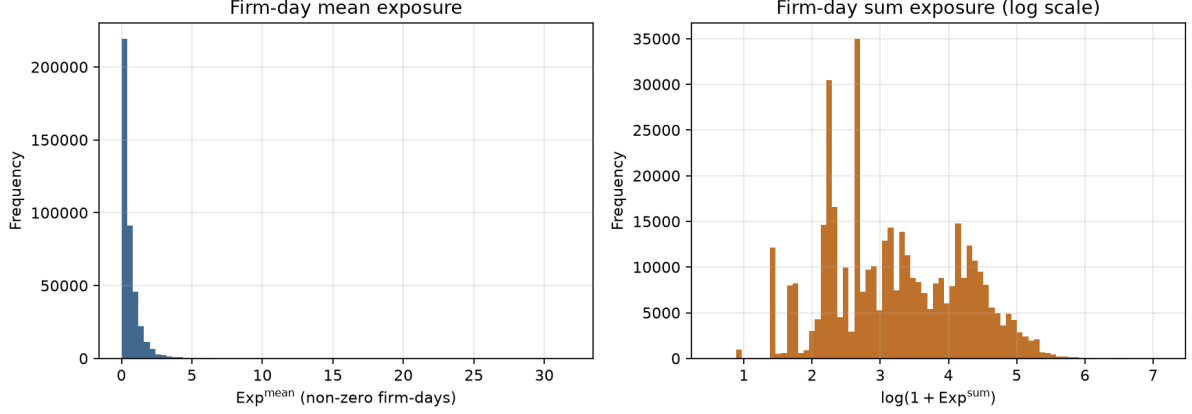


Figure 7: Firm-day exposure distribution. Left panel: $\text{Exp}_{i,t}^{\text{mean}}$ on non-zero firm-days. Right panel: $\log(1 + \text{Exp}_{i,t}^{\text{sum}})$.

E.2 Classifier Validation

The classifier was validated on a held-out spot-check of the first 2,000 (post, sub-industry) judgments. Two qualitative findings shaped the final rubric. First, a default rubric (“score climate misinformation on a continuous 0–100 scale”) over-scored politically charged but non-climate content. Anti-vaccine conspiracy posts, antisemitic content about EU politicians, and electoral-fraud posts that happened to be paired to climate sub-industries by calendar date received high misinformation scores under the default rubric despite making no empirical claims about climate or energy. The headline composite $\text{Exp}_{p,s} = R_{p,s} \cdot M_{p,s}/100$ partially neutralised this leakage because relevance scores on sector-irrelevant content were low, but the misinformation sub-score itself was contaminated. The final rubric (Section 4.3) explicitly restricts misinformation scoring to climate-or-energy claims, and a re-run of the spot-check under the final rubric produced misinformation scores of zero on the politically charged but non-climate items.

Second, sarcastic and parodic content that mocked climate denial received high misinformation scores under the default rubric. A representative example begins “MDPI, the predatory journal of which nothing can be believed, according to Facebook climate experts, recently published a dubious study on anthropogenic warming. . .” and ends with the parody credit “A DONT BELIEVE EVERYTHING YOU READ PRODUCTION”. The default rubric scored this post at $M_{p,s} = 95$ with the type label “denial”; the final rubric, which includes an explicit instruction to treat satire and parody as not-misinformation, scores it at $M_{p,s} = 0$ with the rationale “satirical parody mocking climate denialism, not a sincere assertion of false claims”. Both rubric refinements were committed to the spec before the production run, and the production run used the final rubric.

E.3 Within-Classifier Placebo

Section 6.1.3 reports a within-classifier placebo built from the subset of posts whose per-post sector-relevance score $R_{p,s} < 10$. The placebo aggregates the same exposure measure across only the posts the classifier identified as off-topic for the firm’s sub-industry; under the empirical strategy of Section 5, that aggregate should not move the firm-day outcomes if the headline effect is climate-specific. The placebo coefficients on Models I–III are statistically indistinguishable from zero (Table 5), which supports the climate-specific interpretation without establishing it.

F Heterogeneity and Interaction Checks

F.1 Renewable–Fossil, Regional, and Temporal Splits

Table 26 estimates the three models on the renewable and fossil sub-panels, on each of the five regions, and on the pre/post-2019 samples, using Exp^{mean} under two-way clustering. Figure 8 plots the same coefficients with 95% confidence intervals.

Table 26: Climate-Transition Panel: Coefficient Heterogeneity across Renewable–Fossil, Regional, and Temporal Splits (Two-Way Clustered SE)

Subsample	Model I: Spread%		Model II: CAPM β		Model III: ExcessReturn	
	Coef.	N	Coef.	N	Coef.	N
Renewables	+0.0046	76,680	-0.0012	56,892	-0.0417	56,678
Fossil fuels	+0.0028	391,382	-0.0006	327,125	-0.0180	326,212
United States	+0.0077	219,693	+0.0008	160,380	-0.0102	157,825
United Kingdom	-0.0010	6,954	-0.0014	4,283	+0.0190	4,297
European Union	-0.0005	4,802	+0.0050	4,556	+0.0307	4,678
China	+0.0007	194,377	-0.0016	175,484	-0.0443	176,846
Japan	+0.0034	42,236	-0.0035***	39,314	+0.0085	39,244
Pre-2019	-0.0005	193,062	+0.0014	157,871	-0.0240	164,801
Post-2019	+0.0056**	275,000	-0.0077***	226,146	-0.0039	218,088

Notes: Each cell is the OLS coefficient on Exp^{mean} in the specification of Table 1, estimated on the indicated subsample of the energy panel. Standard errors are clustered on RIC and Date. Cells where $N < 1,000$ are omitted. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

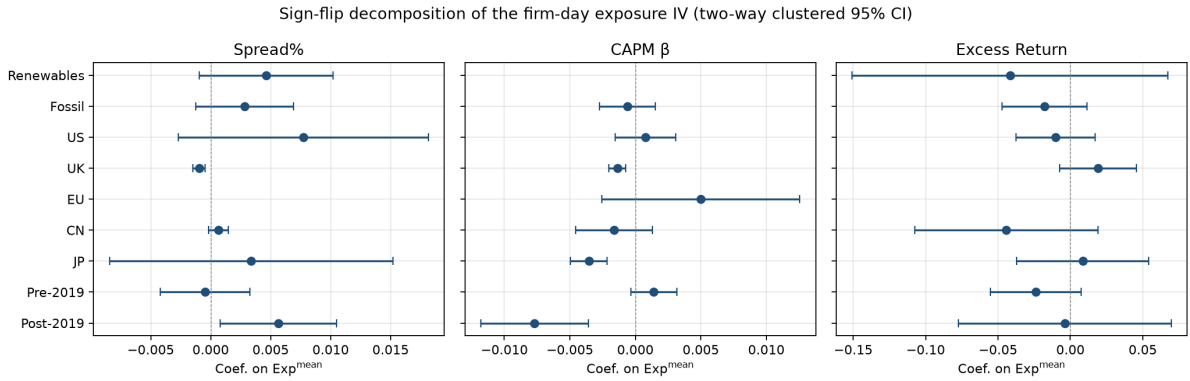


Figure 8: Heterogeneity of the firm-day exposure coefficient across renewable/fossil, regional, and pre/post-2019 splits, by model. Markers are coefficients on $\text{Exp}_{i,t}^{\text{mean}}$; horizontal bars are two-way clustered (RIC + Date) 95% confidence intervals. The post-2019 cell carries the strongest Model I and Model II signals.

The temporal split is discussed in Section 6.1.4; it is reproduced here so that the renewable–fossil, regional, and subperiod estimates can be compared in a common figure.

The regional and renewable–fossil comparisons are less precise. The US Model I coefficient is positive but not significant (+0.0077, $p = 0.15$), while the UK and EU estimates have small samples. The fossil-only and renewable-only Model I coefficients are both positive (+0.0028, $p = 0.17$ and +0.0046, $p = 0.11$), and their difference is not significant. The data therefore do not show that the spread response is cleanly divided between fossil and renewable firms.

F.2 Exploratory Triple Interaction

Table 27 interacts $\text{Exp}_{i,t}^{\text{mean}}$ with Fossil-sub-industry, English-region (US or UK), and Post-2019 indicators in the Model I specification.

Table 27: Triple interaction: Fossil $\times$ English-region $\times$ Post-2019 (Model I, Spread%, two-way clustered SE)

Term	Coef.	SE
Exp^{mean}	+0.0049	(0.0192)
$\text{Exp}^{\text{mean}} \times \mathbb{I}\{\text{Fossil}\}$	-0.0067	(0.0205)
$\text{Exp}^{\text{mean}} \times \mathbb{I}\{\text{English region}\}$	-0.1525*	(0.0862)
$\text{Exp}^{\text{mean}} \times \mathbb{I}\{\text{Post-2019}\}$	-0.0125	(0.0404)
$\text{Exp}^{\text{mean}} \times \text{Fossil} \times \text{English}$	+0.1499*	(0.0870)
$\text{Exp}^{\text{mean}} \times \text{Fossil} \times \text{Post-2019}$	-0.0010	(0.0483)
$\text{Exp}^{\text{mean}} \times \text{English} \times \text{Post-2019}$	+0.3624*	(0.1925)
$\text{Exp}^{\text{mean}} \times \text{Fossil} \times \text{English} \times \text{Post-2019}$	-0.2930	(0.1987)
N	468,062	

Notes: Estimation by Model I; controls and fixed effects as in Table 1. “English region” is US or UK. “Fossil” identifies the seven fossil TRBC sub-industries; “Renewable” firms are the complement within the energy panel. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The three-way interaction is difficult to interpret and is not statistically significant at conventional levels. Two terms are marginal: $\text{Exp} \times \text{English}$ at $p = 0.078$ and $\text{Exp} \times \text{English} \times \text{Post-2019}$ at $p = 0.061$. These estimates are treated as exploratory because the interaction divides the 204-firm sample into small cells and does not identify the coefficients precisely. The simpler subperiod and regional comparisons provide the more interpretable evidence.

G Summary

The appendix consolidates the descriptive and diagnostic evidence behind the paper’s design. The corpus tables document the scale, heterogeneity, and engagement structure of the Facebook data; the network section maps the dissemination structure of the most active post owners. Together they document how the explanatory variable is constructed from variation in post volume, engagement, and timing.

The financial and exposure sections document the breadth and stability of the financial panel, the composition of the climate-transition subsample, and the construction of the firm-day exposure index itself: its distributional properties, the held-out spot-check that fixed the classification rubric before the production run, and the within-classifier placebo. Read as a whole, the appendix connects the competing-liquidity framework of Section 3 to the empirical results in Section 6.